Certified curvature corrections for quantum basket pricing

Sharp error envelopes and cancellation-aware resource comparisons

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Abstract

I study how payoff representation affects the quantum resources required to price capped basket derivatives under a classical no-arbitrage lognormal model. I derive pricing-error certificates from Gaussian conditioning, payoff-boundary curvature and moment bounds, and compare coherent pricing constructions under common information, accuracy and confidence requirements. Across four stressed synthetic contracts, shared-threshold preparation reduces complete logical T-gate upper bounds by 2.34 to 10.00 per cent relative to the cheaper equally optimised implemented rival, after charging state preparation, arithmetic and reflections. In matched Chebyshev diagnostics, numerical conditioning has the lower mean resource charge on the stipulated probability grid. Classical methods certify all 40 prices in the broader test set. The savings apply to the declared finite architecture and precision pool; they do not establish practical quantum advantage.

Keywords: financial mathematics; basket option; Gaussian conditioning; payoff boundary curvature; Gaussian transport; reversible arithmetic; amplitude estimation; quantum finance.

JEL classification: C63, C65, G13.

1 Research question and relation to the literature

I ask whether coherent representations of controlled basket-price corrections reduce complete quantum pricing cost when strong opponents receive the same financial model, analytic controls, derivative constants, moment information, accuracy and confidence. I compare an integrated boundary kernel, a stable two-node quadrature kernel, raw direct second correction, numerical positive first correction, a shared threshold first correction and a common-normal capped-payoff first correction. Quantum mechanics enters through coherent preparation, reversible arithmetic, interference and measurement. The underlying risk-neutral law remains classical, and the priced claim has an explicit no-arbitrage interpretation.

Geometric conditioning and financial smoothing are established (Curran, 1994; Dingeç and Hörmann, 2013; Bayer et al., 2018). Polynomial and conditional Taylor pricing have close antecedents (Ju, 2002; Olivares and Alvarez, 2016); general basket bounds also supply substantive classical competitors (Caldana et al., 2016). Curran's geometric-tail evaluation and approximate below-strike gap calculation already establish the conditioning architecture. [1] The falling log-price derivative operator is already given by Carr (2001). Neither Gaussian call pricing, this operator nor Gaussian quadrature is a new generic mechanism.

Amplitude estimation and amplification provide the established estimation framework (Brassard et al., 2002; Montanaro, 2015). Published quantum derivative-pricing analyses cover signed or unbounded payoffs and substantial implementation costs (Rebentrost et al., 2018; Chakrabarti et al., 2021). In particular, Chen et al. (2026) analyses piecewise-affine payoffs of

[1] I consulted the complete final Curran scan and the final publisher analysis of Olivares–Alvarez, supplemented by its distinct author version where the source ledger identifies formula gaps. Ju's complete article remains unavailable and is not a completed source-level priority exclusion.

linear growth, state preparation, financial grids and tails. [2] Its state-preparation assumptions and elementary-gate complexity are not identical to the realised logical-T contract here. Consequently, an unbounded payoff or a finite error decomposition alone does not establish novelty. Quantum signal-processing pricing and estimation are further published opponents (Stamatopoulos and Zeng, 2024; Rall and Fuller, 2023). I compare a bounded Chebyshev sampling adaptation on all three selected first-correction preparations in each of four cases, with parity-correct circuits, confidence spending and a charged fallback. It does not reproduce every heuristic of published ChebAE, replace the financial payoff arithmetic, or defeat an unimplemented upstream QSP evaluator.

The common-information Gaussian bound follows established convolution and elementary calculus. Common-input sensitivity coupling is longstanding (Glasserman and Yao, 1992), while Chakrabarti et al. (2021) already separates intrinsic and inherited quantum-financial arithmetic errors and propagates derivative-based allowances. Stieltjes pricing and left-limit calculus are established (Marinelli, 2024); its call-spanning measure differs from the bounded increasing payoff measure used here. I therefore claim neither a new generic stability principle nor a new quantum sampling law. The reflection-cost lemma specialises the query and parity constructions of Suzuki et al. (2020); Rall and Fuller (2023). Cross-boundary circuit simplification is established (Stamatopoulos et al., 2020; Carrera Vazquez and Woerner, 2021). [3] The contribution candidate is the exact guarded common-endpoint allocation, its reusable sharp component envelopes and the comparative financial consequences after the same optimisation is supplied to every rival.

The project-specific argument combines sparse payoff-boundary curvature, shared certified norms and positive or signed coherent corrections with a moment-factorised Gaussian transport certificate. The latter integrates derivative envelopes against positive error measures, keeping the small basket-dispersion terms which a global box maximum loses. I derive each part and test their complete composition against stronger evaluators. The general positive-threshold norm theorem is distinguished from the sharper payoff-specific result. The positive second-order cell identity is an exact specialisation of Cerone (2002), Theorem 3, equations (2.13)–(2.15). Neither that identity, its transform nor block refinement is new generic theory. Transform-kernel representations have further published antecedents (Blair et al., 2016).[4] The narrower contribution candidate is the residual-specific derivative envelope and its finite-moment closure, tested with the same information in both transport orders. A close unpublished supplement (Herman et al., 2026) bounds GBM discretisation on uniform physical-coordinate left-endpoint grids. I adapt its first-derivative mechanism to the actual rounded equal-CDF grid; its asymptotic theorem is not a numerical guarantee for these different nodes. The completed common-grid comparison identifies smaller second-order allowances, but almost no value from block refinement. The numerical comparison concerns a frozen compiler, a declared finite precision pool and synthetic models. It can identify conditional implementation gains and adverse cases; it cannot prove a globally optimal quantum algorithm, theoretical priority, a device advantage or empirical exotic-price accuracy. Detailed proofs, consulted source versions and earlier rejected investigations are retained outside the main argument.

2 Financial model and the frozen contracts

Let $\tau > 0$, $r_f \in \mathbb{R}$ and dividend yields $q_i \in \mathbb{R}$ be constant. Under a physical probability, suppose the ex-dividend stock system is

$$dS_i/S_i = b_i\,dt + \Sigma_i\,dW, \tag{1}$$

[2] I verified its published *Mathematical Finance* record; the detailed sections actually consulted are from the authors' May 2026 preprint, as recorded in the source ledger.

[3] The source ledger identifies exact consulted sections and versions. For Carrera Vazquez–Woerner, final APS metadata and abstract were verified, but detailed Section IV was read in arXiv:2005.07711v1, dated 15 May 2020. I do not imply that the inaccessible final theorem text was read.

[4] Only the publisher abstract of Blair et al. was accessible; its full proof-level comparison remains incomplete.

where the filtration is generated by a d_B-dimensional Brownian motion and the constant matrix $\Sigma \in \mathbb{R}^{n\times d_B}$ has full row rank. Choose θ satisfying $\Sigma\theta = b - r_f\mathbf{1} + q$. The exponential density $L_t = \exp(-\theta\cdot W_t - \|\theta\|^2 t/2)$ defines an equivalent measure Q on the finite horizon. Under Q, asset drifts are $r_f - q_i$ and discounted cumulative gains are true martingales. Admissible self-financing discounted wealth bounded below is a supermartingale, precluding arbitrage. Appendix A.9 gives the Gaussian density shift, integrability and localisation proof. [5]

At maturity, independent standard normals $Y \in \mathbb{R}^p$ and $Z \in \mathbb{R}$ give

$$S_i = K\exp(\mu_i + R_i\cdot Y + \lambda Z), \qquad \mu_i = (r_f - q_i)\tau - \tfrac{1}{2}(\|R_i\|^2 + \lambda^2), \tag{2}$$

with $K > 0$ and $\lambda > 0$. The loadings include the horizon scale. Itô's formula for $\log S_i$ proves this expression and gives $\mathbb{E}_Q S_i = Ke^{(r_f-q_i)\tau}$. For positive weights summing to one, define

$$\begin{aligned} A &= \sum_i w_i S_i, & G &= \prod_i S_i^{w_i}, \\ F_H(x) &= \min\{(x-K)_+, H\}, & F_\infty(x) &= (x-K)_+, \\ V &= \mathbb{E}_Q F(A), & \Pi &= e^{-r_f\tau}V. \end{aligned} \tag{3}$$

Here F is the specified capped or ordinary call. Its time-t price is $e^{-r_f(\tau-t)}\mathbb{E}_Q[F(A) \mid \mathcal{F}_t]$. For either payoff $0 \le F(A) \le A$ and $\mathbb{E}A^2 \le \sum_i w_i\mathbb{E}S_i^2 < \infty$ by convexity. The discounted conditional price is therefore a true square-integrable martingale. Adding this claim at that price preserves the same no-arbitrage argument. Boundedness of the raw payoff is unnecessary. [6]

Write

$$\mu_G = \sum_i w_i\mu_i, \quad R_G = \sum_i w_i R_i, \quad d_i = \mu_i - \mu_G, \quad s_i = R_i - R_G, \quad m(y) = \mu_G + R_G\cdot y. \tag{4}$$

Then $\sum_i w_i d_i = 0$ and $\sum_i w_i s_i = 0$ exactly. With

$$T(y) = \sum_i w_i e^{d_i + s_i\cdot y}, \qquad U(y) = T(y) - 1, \tag{5}$$

the identity $A = GT(Y)$ holds. Jensen's inequality gives $T \ge 1$; hence the arithmetic basket exceeds its geometric control pointwise. This deterministic ordering justifies the positive first correction. It does not imply that the ratio and the geometric level are independent.

Every positive-definite maturity log-covariance matrix C admits the common loading

$$\lambda^2 = (\mathbf{1}^\mathsf{T}C^{-1}\mathbf{1})^{-1}, \qquad RR^\mathsf{T} = C - \lambda^2\mathbf{1}\mathbf{1}^\mathsf{T}. \tag{6}$$

The residual matrix is positive semidefinite of rank $n-1$; its factor and an independent common normal reproduce the original covariance and drifts. Appendix A.10 proves these claims. The decomposition establishes model coverage, while numerical cost remains a separate comparison.

The numerical family has $K = 100$, $\tau = 1$, $r_f = 0.03$, $q_i = 0.02$, $n \in \{2, 4, 8\}$, $\lambda \in \{0.05, 0.2, 0.4\}$ and $H \in \{1, 10, 25\}$, giving 27 capped contracts from nine covariance models. I add an ordinary call for each model, giving 36 financial cases. The two-asset anchor uses $w = (1/2, 1/2)$ and $R = (-0.1, 0.2)^\mathsf{T}$. For four and eight assets, the residual loading matrix is a scaled Helmert contrast matrix, with weights $(0.4, 0.3, 0.2, 0.1)$ and $(0.3, 0.2, 0.15, 0.1, 0.08, 0.07, 0.06, 0.04)$ respectively. Its scale gives weighted contrast dispersion 0.0225 in the ideal matrix, matching the two-asset anchor. The actual 70-digit decimal

[5] Full row rank is sufficient for the stated measure construction. If Brownian risk is unspanned, I use the stipulated pricing measure Q; the quantum estimator does not select another measure.

[6] The numerical certificate concerns this single model price. Separately rounded quotes across strikes need not form an arbitrage-free quoted surface.

loadings are frozen as exact rationals, their drifts are computed from those rationals, and exact rational LDL^{T} factorisation checks positive covariance pivots and full volatility row rank. The ideal-to-frozen dispersion discrepancy is below 1.8×10^{-71}. Matching this one dispersion does not match the full payoff distribution or isolate a causal effect of dimension.

These are reproducible synthetic experiments. All figures and tables use synthetic model inputs or generated verification results; no LSEG observations or inaccessible option sample enter the certificates. [7] Historical returns alone would not validate an observed exotic price. The present claims depend on exact model inputs and retained arithmetic certificates, rather than an unreported private empirical sample.

3 Conditional smoothing and the joint derivative certificate

The capped payoff has the positive layer-cake representation $F_H(x) = \int_K^{K+H} \mathbf{1}\{x > k\}\, dk$. Tonelli's theorem therefore gives the conditional kernel

$$P_{\lambda,H}(m) = \mathbb{E}_Z F_H(Ke^{m+\lambda Z}) = \int_0^{\log(1+H/K)} Ke^x \Phi((m-x)/\lambda)\, dx. \tag{7}$$

The undiscounted price is $V = \mathbb{E}_Y P_{\lambda,H}(m(Y) + \log T(Y))$. Conditioning has removed the nonsmooth payoff threshold from the financial integrand. It also supplies the exact geometric control $c_0 = \mathbb{E}P_{\lambda,H}(m(Y))$. The positive first correction $R_1 = P_{\lambda,H}(m + \log T) - P_{\lambda,H}(m)$ is in $[0, H]$. Gaussian smoothing and control identities are useful classical reductions before any quantum encoding is considered (Curran, 1994; Bayer et al., 2018).

The derivative result extends beyond a single capped call. Let ν be a finite nonnegative Borel measure on real log thresholds, of mass H, and define

$$P_\nu(m) = \int \Phi((m-x)/\lambda)\nu(dx), \qquad g_m(t) = P_\nu(m + \log t), \quad t > 0. \tag{8}$$

For the capped call, $\nu(dx) = Ke^x \mathbf{1}_{[0,\log(1+H/K)]}(x)dx$. A cash digital is a point mass; finite positive mixtures of thresholds are also covered. Negative threshold weights, unbounded payoff mass and negative basket weights are not hypotheses of this theorem.

Let $s(r,\ell)$ be signed Stirling numbers of the first kind, with $s(0,0) = 1$ and $s(r+1,\ell) = s(r,\ell-1) - rs(r,\ell)$. The Hermite polynomials satisfy $\mathrm{He}_0 = 1$, $\mathrm{He}_1 = z$ and $\mathrm{He}_{j+1} = z\,\mathrm{He}_j - j\,\mathrm{He}_{j-1}$. Define

$$A_{r,\lambda}(z) = \sum_{\ell=1}^{r} s(r,\ell)\lambda^{-\ell}(-1)^{\ell-1}\,\mathrm{He}_{\ell-1}(z), \qquad D_r(\lambda) = \sup_z |A_{r,\lambda}(z)\phi(z)|. \tag{9}$$

Every polynomial times the Gaussian density is bounded and tends to zero at both infinities. In particular D_r is finite and strictly positive. The separate Hermite norms $\mathcal{H}_j = \sup_z |\,\mathrm{He}_j(z)\phi(z)|$ give the valid triangle bound $D_r \le \sum_{\ell=1}^r |s(r,\ell)|\lambda^{-\ell}\mathcal{H}_{\ell-1}$. Computing the combined norm preserves cancellation between derivative terms which this triangle inequality loses.

Theorem 3.1 (Sharp positive-threshold bound)**.** *For $r \ge 1$, $\lambda > 0$ and the finite positive measure in* (8)*,*

$$g_m^{(r)}(t) = t^{-r}\int A_{r,\lambda}((m+\log t-x)/\lambda)\phi((m+\log t-x)/\lambda)\nu(dx), \qquad |g_m^{(r)}(t)| \le HD_r(\lambda)t^{-r}. \tag{10}$$

The constant HD_r is the smallest uniform constant over measures of mass H, real m and positive t.

[7] Any future LSEG calibration is restricted to my authorised individual-use entitlement. Extraction and analysis code, together with only the permitted limited academic, non-commercial summaries, may be released. Underlying observations and systematic derived datasets remain excluded from GitHub and other third-party redistribution under the supplied institutional guidance.

The falling log-price derivative operator produces the polynomial in (9); positivity bounds its integral. A point-mass threshold attains the resulting operator norm. Appendix A.2 gives the full differentiation and sharpness proof.

This sharpness concerns the stated measure class; a particular capped-call measure or basket remainder need not attain the norm. The proof also identifies exactly which object must be computed. For rational λ, $Q_{r,\lambda} = A'_{r,\lambda} - zA_{r,\lambda}$ is a rational polynomial of degree r. Every positive absolute maximum of $A_{r,\lambda}\phi$ lies at a real root of $Q_{r,\lambda}$. Exact square-free reduction, a rational root-counting sequence and finite rational isolation find every distinct root; Gaussian evaluations over the isolating intervals give an outward upper certificate $\overline{D}_r \geq D_r$. Appendix B supplies the termination and enclosure argument. All finite calculations below use $\overline{D}_r$, rather than an uncertified maximisation on a numerical mesh. Their arithmetic is implemented with Arb midpoint-radius enclosures (Johansson, 2017).

The separate Hermite and Fourier triangle norms remain competent comparators. They distinguish cancellation gains from tighter individual derivative bounds; Appendix A.13 gives their recurrences.

4 Exact Gaussian coefficients and finite classical screening

Put $\sigma^2 = \lambda^2 + \|R_G\|^2$. For a multi-index $\gamma \in \mathbb{N}_0^n$, define $|\gamma| = \sum_i \gamma_i$, $\gamma! = \prod_i \gamma_i!$, $w^\gamma = \prod_i w_i^{\gamma_i}$, $d_\gamma = \sum_i \gamma_i d_i$ and $v_\gamma = \sum_i \gamma_i s_i$. The multinomial theorem and completion of the Gaussian square imply

$$T(y)^j = \sum_{|\gamma|=j} \frac{j!}{\gamma!} w^\gamma e^{d_\gamma + v_\gamma \cdot y}, \tag{11}$$

$$\mathbb{E}[e^{v\cdot Y} P^{(\ell)}_{\lambda,H}(\mu_G + R_G \cdot Y)] = e^{\|v\|^2/2} P^{(\ell)}_{\sigma,H}(\mu_G + R_G \cdot v). \tag{12}$$

The square-completion and dominated-differentiation argument is given in Appendix A.11.

For ordinary calls the same tilt formula follows from the Gaussian exponential moment and dominated differentiation on compact argument intervals. In the second-correction study I need only

$$c_1 = \sum_i w_i e^{d_i + \|s_i\|^2/2} P'_\sigma(\mu_G + R_G \cdot s_i) - P'_\sigma(\mu_G), \tag{13}$$

where P_σ is the chosen capped or vanilla kernel. The finite positive-threshold theorem below remains a capped-payoff result; the vanilla remainder uses the boundary theorem in Section 5. The shift $R_G \cdot v$ retains the covariance between the geometric level and the ratio. Removing it generally changes the coefficient.

Let

$$J_{j,\ell} = \sum_{|\gamma|=j} \frac{j!}{\gamma!} w^\gamma e^{d_\gamma + \|v_\gamma\|^2/2} P^{(\ell)}_{\sigma,H}(\mu_G + R_G \cdot v_\gamma), \tag{14}$$

$$\mathcal{M}_r = \sum_{j=0}^{r} (-1)^{r-j} \binom{r}{j} \sum_{|\gamma|=j} \frac{j!}{\gamma!} w^\gamma e^{d_\gamma + \|v_\gamma\|^2/2}. \tag{15}$$

Then $\mathcal{M}_r = \mathbb{E}U^r$ and all terms are finite. Taylor's theorem about $t = 1$ defines

$$c_0 = P_{\sigma,H}(\mu_G), \qquad c_a = \frac{1}{a!} \sum_{\ell=1}^{a} s(a,\ell) \sum_{j=0}^{a} (-1)^{a-j} \binom{a}{j} J_{j,\ell}, \quad a \geq 1. \tag{16}$$

The inner alternating sum is the exact expectation of $U^a P^{(\ell)}_{\lambda,H}(m(Y))$. Its cancellation is kept inside outward arithmetic, not estimated from printed decimal precision.

Theorem 4.1 (Finite Gaussian price certificate)**.** *Under* (2)–(5)*, for every finite integer* $k \geq 0$*,*

$$\left| V - \sum_{a=0}^{k} c_a \right| \leq \frac{H\overline{D}_{k+1}(\lambda)\mathcal{M}_{k+1}}{(k+1)!}. \tag{17}$$

Each coefficient and moment is evaluated by the finite sums above. For rational inputs and positive common scale, validated enclosures of all endpoints exist at any prescribed positive arithmetic accuracy.

The integral Taylor remainder and Theorem 3.1 give the displayed radius. Gaussian exponential moments make every finite-order coefficient and remainder integrable. Appendix A.3 supplies the full argument.

For ordinary calls and finite $k \geq 1$, write $C_r = \prod_{j=0}^{r-1}(\partial_m - j)P_{\lambda,\infty}$ and take $M_r \geq \|C_r\|_\infty$ for $r \geq 2$. The same Gaussian tilts evaluate $c_a = \mathbb{E}[U^a C_a(m)]/a!$ and yield

$$\left| V - \sum_{a=0}^{k} c_a \right| \leq M_{k+1}\mathbb{E}U^{k+1}/(k+1)!. \tag{18}$$

Indeed, $g_m^{(k+1)}(t) = t^{-k-1}C_{k+1}(m + \log t)$ is bounded by M_{k+1} for $t \geq 1$. The integral Taylor remainder and finite Gaussian moments prove the inequality exactly as above. The unbounded P, P' terms remain integrable under each Gaussian tilt; the higher falling derivatives have bounded Gaussian boundary kernels. At $k = 1$ positivity of the ordinary call curvature gives the one-sided interval in Corollary 5.2. This is a finite-order classical competitor, with no infinite-series convergence claim.

If $[L_k, U_k]$ encloses the coefficient sum and $\overline{R}_k$ bounds the right side, I use $[L_k - \overline{R}_k, U_k + \overline{R}_k] \cap [0, H]$. Intersections across finitely many retained degrees can only improve this interval. I stop when its midpoint radius meets the monetary target, and otherwise use a separately certified method or declare failure. The argument is finite-order: finite lognormal moments do not establish convergence of the infinite Taylor expansion. The earlier Fourier triangle remainder eventually grows for any nonzero centred loading, so taking ever more terms is not a valid convergence policy.

The kernel evaluations are explicit. With $L_H = \log(1 + H/K)$, $x_0 = m/\sigma$, $x_1 = (m - L_H)/\sigma$ and $z_i = x_i + \sigma$,

$$P_{\sigma,H}(m) = K\{e^{m+\sigma^2/2}[\Phi(z_0) - \Phi(z_1)] - \Phi(x_0) + (1 + H/K)\Phi(x_1)\}. \tag{19}$$

Independent derivative-series and strike-integral checks verify the kernel evaluations; Appendix A.12 gives the cancellation identities and differentiation details.

The certificate extends to unequal positive spots by replacing d_i with $\log(S_{i,0}/G_0) + \mu_i - \mu_G$, where $G_0 = \prod_i S_{i,0}^{w_i}$, and shifting every scalar-kernel argument by $\log(G_0/K)$. Weighted centring and the covariance shift survive. Positive asset-monitor observations can likewise be treated as basket components when their joint Gaussian log prices admit an equal independent common loading $\lambda > 0$. The component count then includes monitoring times; the multinomial cost can grow substantially. These are analytical extensions, not additional empirical tests.

5 Payoff curvature and coherent correction evaluators

Let $P_F(m) = \mathbb{E}F(S_{\text{ref}}e^{m+\lambda Z})$, $D = \partial_m$ and $C_r = D(D-1)\cdots(D-r+1)P_F$. I write $m = m(Y)$ and $U = U(Y)$. Taylor's integral identity gives the exact correction

$$\begin{aligned} R_2 &= P_F(m + \log(1+U)) - P_F(m) - UP_F'(m) \\ &= \frac{U^2}{2}\mathbb{E}_{Z_B}[(1 + Z_B U)^{-2}C_2(m + \log(1 + Z_B U))]. \end{aligned} \tag{20}$$

where $Z_B \sim \mathrm{Beta}(1,2)$. The zero extension at $U = 0$ requires no division. Analytic evaluation of the payoff boundary removes the strike auxiliary variable from the earlier integral construction.

Theorem 5.1 (Sparse payoff-boundary correction)**.** *Suppose* $F(s) = a + bs + \sum_{j=1}^{J} \alpha_j (s - K_j)_+$ *with finite* J *and positive* $K_j, S_{\mathrm{ref}}, \lambda$*. Then*

$$C_2(m) = \sum_{j=1}^{J} \frac{\alpha_j K_j}{\lambda} \phi\left(\frac{m - \log(K_j/S_{\mathrm{ref}})}{\lambda}\right). \tag{21}$$

If $\alpha_j \geq 0$ *for all* j*, then* $R_2 \geq 0$ *for every* $U \geq 0$*. For all finite signed kink weights,* $|R_2| \leq M_2 U^2/2$ *whenever* $M_2 \geq \|C_2\|_\infty$*.*

Each payoff kink contributes a Gaussian boundary density to $P_F'' - P_F'$; the affine payoff terms vanish under $D(D-1)$. The nonnegative Taylor weight gives the sign under convexity and the absolute norm bound for signed kink weights. Appendix A.4 proves every step.

Adding $a + bs$ to the payoff leaves R_2 exactly unchanged: its conditional $g_m(t) = a + bS_{\mathrm{ref}} e^{m+\lambda^2/2} t$ is affine in t and has zero second remainder. Calls and puts at the same strike therefore share the ideal curvature correction; the deterministic coefficients restore their established discounted difference $e^{-r_f \tau}(\mathbb{E}A - K)$. This is an applicability consequence of the derivation, rather than a new put–call parity claim or an additional numerically tested payoff family.

Corollary 5.2 (Continuous convex screening)**.** *For a convex finite-kink payoff under* (2)*, suppose* $M_2 \geq \|C_2\|_\infty$*. Then, in every retained asset dimension,*

$$c_0 + c_1 \leq V \leq c_0 + c_1 + M_2 \mathbb{E}U^2/2. \tag{22}$$

Taking expectations of the positive bounded residual gives the interval, with Gaussian tilts evaluating its coefficients. Appendix A.5 gives the complete proof. This analytical screen precedes financial discretisation and quantum estimation. It does not require an exponentially large product grid or assume that a small correction deserves a quantum computation.

For the cap, set $c = 1 + H/K$, $\ell = \log c$ and $z = m/\lambda$; then

$$C_2(m) = \frac{K}{\lambda}\{\phi(z) - c\phi(z - \ell/\lambda)\}. \tag{23}$$

For the vanilla call $C_2 = K\phi(m/\lambda)/\lambda \geq 0$ and its exact norm is $K/(\lambda\sqrt{2\pi})$. I use a rational outward upper endpoint M_2 in every graph. For the cap, the two stationary points in Appendix G certify its exact global norm. I also certify the unique global peak of P' as M_1. At $\lambda = 0.05, H = 25$, the resulting upper constants are approximately 39.23 and 54.99 per cent of $H\overline{D}_2$ and $H\overline{D}_1$, respectively. The first-derivative comparator also has the valid bound $\min\{K+H, H\overline{D}_1\}$; against that stronger bound, its reduction at this case is approximately 12.25 per cent. These constants are supplied to every current coherent opponent and the new classical evaluators; they are not reserved for the proposed oracle.

On the realised financial support, $0 \leq U \leq U_*$. For $U_* > 0$, define

$$\begin{aligned} B_2 &= M_2 U_*^2/2, \qquad t = 1 + z_B U, \\ a &= (U/U_*)^2 t^{-2}, \quad h = (1 + C_2(m + \log t)/M_2)/2. \end{aligned} \tag{24}$$

For capped calls, the boundary evaluator prepares

$$p^{\mathrm{boundary}} = (1-a)/2 + ah, \quad X = 2p^{\mathrm{boundary}} - 1, \quad B_2 X = U^2 t^{-2} C_2(m + \log t)/2. \tag{25}$$

All factors lie in $[0,1]$ and $|X| \leq U^2/U_*^2$. For vanilla calls the native positive encoding is $p^{\mathrm{boundary}} = a\gamma(m + \log t)$ with $\gamma = C_2/M_2 \in [0,1]$, and its monetary decoder is $B_2 p$, rather

than $B_2(2p-1)$. The raw payoff is unbounded, but this encoded correction is bounded on the certified finite financial support. Gaussian tails remain a separate financial obligation.

A strong deterministic evaluator uses the two-node Gauss-Jacobi rule for $2(1-z)dz$:

$$z_\mp = (4 \mp \sqrt{6})/10, \quad w_\mp = \tfrac{1}{2} \pm 1/(3\sqrt{6}), \quad R_2^{\mathrm{GJ}} = \frac{U^2}{2} \sum_{\pm} w_\pm (1 + z_\pm U)^{-2} C_2(m + \log(1 + z_\pm U)). \tag{26}$$

It preserves the same small-factor representation without a sampled auxiliary or raw price subtraction. I implement its exact dyadic constants, with complementary positive weights summing to one, and charge the algebraic node and weight discrepancies. A fourth-order Hermite remainder proves its monetary quadrature bound, not a heuristic comparison of two quadrature values.

The raw direct opponent uses the same $c_0 + c_1$, M_2 and B_2:

$$p^{\mathrm{raw}} = \tfrac{1}{2} + R_2/(2B_2) \tag{27}$$

for the cap and $p^{\mathrm{raw}} = R_2/B_2$ for vanilla calls. Its arithmetic errors are propagated before cancellation. The positive first-correction opponent uses c_0, $R_1 = P(m + \log T) - P(m)$ and

$$p^{(1)} = R_1/B_1, \qquad B_1 = \min\{H, M_1 U_*\} \tag{28}$$

for the cap. For vanilla calls $B_1 = M_{1,\mathrm{finite}} U_*$, where $M_{1,\mathrm{finite}} = K e^{m_+ + \lambda^2/2}$ bounds the conditional forward on its certified argument interval. There is no globally bounded vanilla first log-price derivative. Every scalar, normaliser and input-domain certificate is method-specific, with identical information sharing.

Proposition 5.3 (Auxiliary variance penalty)**.** *With continuous auxiliaries and the same finite financial law, the exact boundary score has conditional mean $R_2(Y)$ and $\mathrm{Var}(B_2 X) = \mathrm{Var}(R_2(Y)) + \mathbb{E}\,\mathrm{Var}(B_2 X \mid Y)$ for the capped encoding; the analogous identity uses $B_2 p$ for the vanilla encoding.*

The conditional mean is $R_2(Y)$, so the law of total variance separates financial dispersion from auxiliary randomisation. The full square-expansion argument is in Appendix A.6. This established variance decomposition prevents attributing an intrinsic variance reduction to auxiliary randomisation. Any gain must survive its actual evaluator and estimation costs. The stable quadrature opponent is an implemented comparator; the finite pool is not a lower bound on every possible cancellation-preserving pricing method.

6 Finite preparation and complete monetary error

Independent uniform labels prepare a realised truncated-normal quantile law. With $L > 0$, $N = 2^{b_y}$ and $\alpha_1 = \Phi(L) - \Phi(-L)$, I use

$$y_j = \Phi^{-1}(\Phi(-L) + \alpha_1(j + 1/2)/N), \quad \widehat{y}_j = 2^{-q}\,\mathrm{round}(2^q y_j), \quad \alpha = \alpha_1^p. \tag{29}$$

Outward CDF brackets prove unique rounding, order and domain membership. Duplicate coordinates keep their label multiplicities. Hadamards prepare labels; coherent lookup computes coordinates and its inverse clears work. If $[\ell_y, h_y]$ covers every ideal enclosure and actual node, then

$$U_* \geq \max_{v \in \{\ell_y, h_y\}^p} \sum_i w_i e^{d_i + s_i \cdot v} - 1 \tag{30}$$

is sufficient: the sum is convex in each coordinate, so maximisation can move successively to endpoints. Outward vertex evaluation certifies all nine model bounds. The continuous tail box and finite normalisation support have different roles.

The boundary method samples the exact cell-mass midpoint Beta law. The minimum of two b_Z-bit uniform labels has mass $2h_Z(1-z_j)$ at $z_j=(j+1/2)h_Z$, $h_Z=2^{-b_Z}$, which equals the continuous Beta$(1,2)$ mass of that cell. No strike register is needed. The stable method instead uses the two exact dyadic Gauss constants. Let $M_{r,N}=\mathbb{E}_N U^r$ under the actual financial label law. Appendix D proves, before the external financial factor,

$$\beta_{\mathrm{B}} \le h_Z^2(M_3M_{3,N}/12+M_4M_{4,N}/48), \tag{31}$$

$$\beta_{\mathrm{GJ}} \le M_6M_{6,N}/14400+\frac{M_3M_{3,N}}{2}\sum_{\pm} w_{\pm}\delta z_{\pm}+\frac{M_2M_{2,N}}{2}\sum_{\pm}\delta w_{\pm}. \tag{32}$$

Here $M_r \ge \|C_r\|_\infty$ is a derivative bound, distinct from the moment $M_{r,N}$. The common norm ledger uses $C_{r+1}=(D-r)C_r$, exact polynomial stationary-root isolation and outward Gaussian evaluation. Actual fourth and sixth moments strengthen the stable opponent; finite-support fallbacks remain valid but are not used to weaken it. Raw direct and first-correction methods have zero auxiliary error.

Every function is implemented by a one-dimensional endpoint table with outward nodal enclosures and signed-magnitude interpolation. The graph propagates absolute errors and implemented ranges through exponentials, positive ratio arithmetic, logarithms, inverse squares and the specified payoff kernels. Direct subtraction receives no assumed favourable relative error. Domain violations, overflow and ambiguous signed ranges reject a graph. Appendix C states the interpolation and arithmetic contracts; the supporting source retains every node. Computing a numerical threshold, toggling a comparator flag and applying the literal inverse cleans private work. For integer $k\in\{0,\dots,2^q\}$ and uniform $j\in\{0,\dots,2^q-1\}$, the predicate $j<k$ gives exact Born success probability $k/2^q$, including both endpoints. Complete emitted preparations replace any free-oracle assumption.

Appendix E derives the price-independent financial certificate. A local signed-cell calculation supplies the sharper two-asset bounds. For higher dimensions, I express the cell error as a positive measure, compute its exponential moments and use tensor hybrids to integrate dispersion-weighted derivative envelopes. All 36 first- and second-correction targets now have complete financial transfers. The global transport family retains its three insufficient two-asset bounds; I use the stronger local certificate in those cases. The construction is classical setup, with explicit operation counts and outward errors.

Theorem 6.1 (Complete monetary error)**.** *Let e_c bound the chosen coefficient midpoint error, let $[F_-,F_+]$ enclose the external mass with midpoint and radius e_F, and let the continuous-to-finite correction discrepancy be at most β_Y. Suppose the ideal and implemented probabilities differ by at most η_p and the implemented score has absolute value at most one. On the jointly certified estimation event,*

$$|\widehat{V}-V| \le e_c+\beta_Y+F_+(\beta_{\mathrm{aux}}+dB\eta_p+e_{\mathrm{stat}})+e_FB, \tag{33}$$

where $(B,d)=(B_2,2)$ for a capped centred second correction and $(B,d)=(B_2,1)$ for a native positive second correction, or $(B_1,1)$ for the first correction. The auxiliary term is chosen from (31) *or* (32)*, or is zero for a raw opponent. Statistical error is monetary before the mass, at its separately certified joint confidence. Discounting multiplies the bound by $e^{-r_f\tau}$.*

Each deterministic replacement is charged before the bounded probability decoder is applied. The decoder contributes $dB\eta_p$, and uncertain external mass contributes e_FB on the joint estimation event. Appendix A.7 gives the full decomposition.

A graph is admitted by the complete monetary budget in Theorem 6.1, rather than by its number of fractional bits.

7 Signed flags and competent finite moment priors

The centred encoding can have probability near one half even when its monetary correction is small. I therefore also implement two exact signed parts of the represented probability:

$$\widehat{p}_+ = \max(0, 2\widehat{p} - 1), \qquad \widehat{p}_- = \max(0, 1 - 2\widehat{p}), \qquad \widehat{p}_+ - \widehat{p}_- = 2\widehat{p} - 1. \tag{34}$$

The transformations use exact dyadic scaling and positive subtraction. They preserve the difference on every represented basis input, including $\widehat{p} \in \{0, 1/2, 1\}$. They are separate success flags with separately charged preparation and confidence costs; I do not estimate a fictitious flag with the variance of a signed classical output. Their deterministic recombination uses the centred base certificate in Theorem 6.1, avoiding a double charge for an exact integer identity.

The retained truncated-normal quantile law has $\omega = 1$ and external mass $\alpha = \alpha_1^p$. For the uniform-box variant on $[-L, L]^p$, $\omega(y) = e^{-\|y\|^2/2}$ and $C_L = (2L/\sqrt{2\pi})^p$. These weights belong to the prepared score; their monetary factors are decoded after estimation. The uniform-box variant prepares $p_\omega = (1 + X_\omega)/2$, where $X_\omega = \omega X$, and forms the signed parts of its represented probability in (34). The quantile variant has $X_\omega = X$. For each capped construction, $p_\pm \le |X_\omega| \le \omega U^2/U_*^2$. The positive-part map is one-Lipschitz, so the implemented flags satisfy

$$\mathbb{E}_N \widehat{p}_\pm \le u_2 := \min\{1, M_{2,N}/U_*^2 + 2\eta_p\}. \tag{35}$$

The expectation is under the prepared label law. Neither its Gaussian box mass α nor its uniform-box monetary factor C_L belongs in this probability prior. They enter monetary reconstruction after estimation. A prior reduction improves a canonical estimator's amplitude error bound only if the increased flag-preparation costs and two confidence allocations are also included.

For the native positive vanilla correction the tighter implemented prior is $u_2 = \min\{1, M_{2,N}/U_*^2 + \eta_p\}$, since its decoder has no factor two. These bounds also apply to the stable rule: its positive weights sum one and preserve the same pointwise envelope.

The finite moments are exactly computable without a financial price. Define the one-dimensional exponential sum

$$\mathcal{L}_N(v) = N^{-1} \sum_{j=0}^{N-1} e^{v\widehat{y}_j}, \tag{36}$$

and, for uniform-box weighting, replace each summand by $e^{vy_j - y_j^2/2}$. Independence and the multinomial expansion give

$$\mathbb{E}_N[\omega T^r] = \sum_{|\gamma|=r} \frac{r!}{\gamma!} w^\gamma e^{d\gamma} \prod_{k=1}^p \mathcal{L}_N\left(\sum_i \gamma_i s_{ik}\right), \qquad M_{r,N} = \sum_{j=0}^r (-1)^{r-j} \binom{r}{j} \mathbb{E}_N[\omega T^j]. \tag{37}$$

These are finite sums over actual dyadic nodes. For the weighted law the zeroth raw moment is $\mathbb{E}_N\omega$, not one. All cancellations remain enclosed outwards. Thus a moment calculation can use products of one-dimensional sums instead of N^p financial price evaluations. Its scalar preprocessing cost is still retained; separability does not make it free.

The positive first-correction comparator also receives a competent prior. Both tested payoffs are increasing and one-Lipschitz in the underlying price. For fixed $Y = y$, $A = GT$ and the independent common factor therefore imply

$$0 \le R_1(y) \le K e^{m(y) + \lambda^2/2} U(y). \tag{38}$$

The corresponding finite expectation retains the level-ratio correlation:

$$\Gamma_N = K e^{\mu_G + \lambda^2/2} \left\{ \sum_i w_i e^{d_i} \prod_{k=1}^p \mathcal{L}_N(R_{G,k} + s_{ik}) - \prod_{k=1}^p \mathcal{L}_N(R_{G,k}) \right\}. \tag{39}$$

Expanding $e^{R_G \cdot Y}(T-1)$ proves this identity. Combining it with $R_1 \le H$ and $R_1 \le M_1 U$ gives

$$u_1 = \min\left\{1, \frac{\min\{H\mathbb{E}_N\omega, M_1 M_{1,N}, \overline{\Gamma}_N\}}{B_1} + \eta_{p,1}\right\}, \tag{40}$$

where $\overline{\Gamma}_N$ is an outward upper endpoint. For vanilla calls the mean bound is $\overline{\Gamma}_N$; I omit the finite-cap bounds rather than assuming a globally bounded P'. For the three one-dimensional grids, I also enclose $M_{4,N}$ and $M_{6,N}$ by direct nonnegative summation over the actual 1,024 nodes, charging its 2,048 conditional exponential evaluations per model. Replacing $\mathbb{E}(e^{R_G \cdot Y}U)$ by a product of separate expectations would generally invalidate this bound. The prior is derived from moments and the Lipschitz property, not a secretly supplied option price.

The first-correction comparator also receives the Gaussian coefficient c_1, already supplied to every second-correction method. Define $B_Q = M_2 \mathbb{E}_Q U^2/2$. For a cap, $\mathbb{E}_Q R_1 = c_1 + \mathbb{E}_Q R_2 \in [c_{1,-} - B_Q, c_{1,+} + B_Q]$; for an ordinary call the lower endpoint is $c_{1,-}$, since $R_2 \ge 0$. If this continuous interval is $[L_Q, U_Q]$ and the first-residual transfer is β_1, with $0 < \alpha_- \le \alpha \le \alpha_+$, then

$$L_N = \frac{\max\{0, L_Q - \beta_1\}}{\alpha_+}, \qquad U_N = \min\{B_1, (U_Q + \beta_1)/\alpha_-, U_{N,\text{previous}}\}. \tag{41}$$

These endpoints enclose the exact finite mean: insert $|\mathbb{E}_Q R_1 - \alpha \mathbb{E}_N R_1| \le \beta_1$, use positivity and divide by the adverse mass endpoints. Intersecting the earlier upper bound is valid. The implemented probability mean therefore belongs to

$$[\ell, u] = [\max\{0, L_N/B_1 - \eta_p\}, \min\{1, U_N/B_1 + \eta_p, u_{1,\text{previous}}\}]. \tag{42}$$

Its midpoint gives an analytical stop when $\alpha_+ B_1(u-\ell)/2$ plus the full deterministic allowance meets the target. Otherwise u supplies the canonical Bernoulli variance bound. The full Gaussian second moment is another finite exponential sum; no basket-price answer enters this information refinement.

Before any method estimates a correction, I also apply the same continuous low-order price screen. The capped interval is $[c_0 + c_1 - B_Q, c_0 + c_1 + B_Q] \cap [0, H]$; the ordinary-call interval is $[c_0 + c_1, c_0 + c_1 + B_Q] \cap [0, \infty)$, with all coefficient endpoints outward. The global remainder theorem proves containment and its rational midpoint radius proves the stop. A sufficient continuous interval requires no grid or quantum circuit, so every competing method receives this same decision. Higher-order classical pricing remains a separately reported practical opponent. The supporting proof retains all interval and decoder steps of the information refinement.

8 Estimation and complete logical resource accounting

Let $\mathcal{A}$ be a clean preparation with success probability p, including every financial and comparator register. The Grover iterate is formed from $\mathcal{A}$, its inverse and the good-state and zero-state reflections. Canonical phase estimation uses controlled powers of this iterate (Brassard et al., 2002); I retain the controlled global signs of the reflections, since a global sign becomes a relative phase under control. For $M = 2^m$, the standard amplitude estimate obeys, on its success event,

$$|\widetilde{p} - p| \le \frac{2\pi\sqrt{p(1-p)}}{M} + \frac{\pi^2}{M^2}. \tag{43}$$

Appendix A.14 derives the bound from the two Grover eigenphases and the phase-grid displacement. The retained confidence calculation supplies its certified success probability.

If $p \le u$, the variance term is bounded by $u(1-u)$ for $u \le 1/2$ and by $1/4$ otherwise. I apply (35) or (40), rather than assuming an arbitrary favourable amplitude. Exact finite postprocessing error and the compiled inverse Fourier transform's implementation error are

charged separately. Fresh median repetitions amplify confidence by the exact binomial tail for their certified single-shot success probability. The allocation is fixed before observing results. Two signed-part estimates share a union budget; at most 36 prespecified prices share an overall failure allowance of one per cent. The retained protocol assigns 1/3600 per selected price. [8] Deterministic selection among certified implementations requires no statistical selection penalty because no measurement outcomes are used to choose their resource plans.

The inverse Fourier transform uses explicitly certified Clifford and T sequences (Ross and Selinger, 2016). Its retained and omitted phase errors are charged in every complete plan. Appendix A.15 gives the synthesis details and largest-width diagnostics.

A complete count includes the initial preparation, every controlled financial preparation and inverse, good and zero reflections, inverse Fourier transform, confidence repetitions, observed qubits and reset of reused shot registers. Each method receives both coherent lookup architectures and can select its lowest complete cap before any estimator shots. The study retains both exact seven-T and coherent eight-T construction profiles and their emitted chronological tapes. QROM mechanisms have established predecessors (Babbush et al., 2018); using them correctly is an implementation obligation, not a novelty claim. An exact lookup compute/erase pair costs 14 T gates, while the coherent AND pair costs eight; ordinary arithmetic uses literal seven-T Toffolis in both modes. Financial table generation, norm generation, finite moment sums and financial-grid certification are reported as classical setup. A logical T count is not directly equated to CPU time or fault-tolerant wall-clock time.

For a centred second-correction flag with probability tolerance e_p, the monetary statistical allowance is $2B_2e_p$ before the factor. For two signed flags with tolerances e_+, e_- it is $B_2(e_+ + e_-)$, at their joint confidence. The positive first correction uses B_1e_p, and the native positive vanilla second correction uses B_2e_p. I select only configurations satisfying the complete monetary constraint, with at most half the requested radius assigned to deterministic error. The final study fixes $q = 28$, $b_y = 10$ and the remaining Beta width at eight, then tunes each scalar table independently within six to twelve address bits. The unchanged all-twelve program remains in every method pool, alongside three greedy searches with deterministic allowances 0.001, 0.0025 and 0.005. The search ranks certified cell savings per marginal monetary error, with a fixed node-name tie rule; the final choice minimises the actual complete T cap. Every proposed map, guard result, chosen step and cost is retained. I also admit an exact integer constant-propagation pass and the unchanged emitter equally for all methods. The pass preserves represented outputs, widths and all input registers. It produces no rewrite in the 204 higher-dimensional production graph structures; its small-fixture simplifications are not extrapolated into financial resource savings. This is a reproducible finite search, rather than a global precision optimum. Equal financial accuracy, information, confidence, architecture and admissibility are the matched requirements.

9 Retained findings and comparison boundary

The earlier correction-family study retains 502 candidate programs and 141,676 recorded allocation decisions; later registered pools are reported separately below. The higher-dimensional pool adds 329 candidates, 117,871 genuine precision proposals and 33 infeasible starting allocations. The combined record preserves every earlier implementation and strengthens 113 first-correction information intervals. Both coherent architectures receive the same controls, financial accuracy and selection policy. The family confidence allocation remains 1/3600 per selected price, with a union allowance of one per cent across all 36 cases.

Table 1 identifies the resolved financial gap. The nominal product grid contains 2^{70} labels in seven dimensions, but the transfer construction evaluates finite exponential-polynomial moments, not that Cartesian set. The derivative bounds retain U, T_j and T_{jj} moments, actual

[8] The four-case stressed comparison is reported separately from the original 36-contract family, using the same per-price allowance.

rounded nodes, Gaussian tails and all geometric-loading terms. The loadings R_G are nonzero in both higher-dimensional families; weighted centring of s_i does not justify discarding them.

Assets	Outer dimension	Nominal labels	Max. $\beta_{Y,2}$	Max. $\beta_{Y,1}$	Cases
2	1	2^{10}	0.000937	0.001168	12
4	3	2^{30}	0.002284	0.004914	12
8	7	2^{70}	0.001107	0.003572	12

Table 1: Complete continuous-to-finite financial allowances, rounded upwards, under unchanged $L = 6$, $b_y = 10$, $q = 28$. The target allowance is 0.005. Two-asset rows use the sharper local cell certificate; higher-dimensional rows use positive transport moments. Arithmetic, auxiliary and statistical errors are additional terms, not included in these financial allowances. The table makes the distinction between nominal grid size and executed certificate work explicit.

Shared continuous second-order intervals resolve 18 contracts before quantum estimation. In the remaining 18 active comparisons, the best second correction beats the informed positive first correction in ten cases, with reductions of approximately 13.9 to 92.6 per cent. Table 2 displays the complete-cap ratios for the entire family; it reveals adverse cases as well as improvements. Matching one dispersion quantity across models does not isolate a causal dimension effect.

Assets	λ	$H = 1$	$H = 10$	$H = 25$	Ordinary call
2	0.05	3.071_{D}	3.022_{D}	1.700_{D}	0.349_{D}
2	0.2	A	0.861_{D}	0.689_{D}	0.088_{D}
2	0.4	A	A	0.355_{D}	A
4	0.05	2.595_{D}	1.323_{D}	1.310_{D}	0.261_{D}
4	0.2	A	A	0.366_{D}	0.074_{D}
4	0.4	A	A	A	A
8	0.05	A	2.238_{D}	1.205_{D}	0.240_{B}
8	0.2	A	A	0.584_{D}	A
8	0.4	A	A	A	A

Table 2: Lowest second-correction complete T cap divided by the informed first-correction cap, rounded to three decimals. D denotes raw direct R_2, B the boundary oracle and J stable Gauss-Jacobi; J is never the lowest active second correction here. A denotes the shared analytical price interval, with zero quantum calls for every method. A ratio below one is a conditional logical-resource improvement, with the same monetary radius 0.01, confidence, architectures and precision-pool policy. Exact caps, selected maps and full budgets remain in the archive.

Raw direct evaluation is the cheapest second-correction implementation in 17 active cases. For the eight-asset ordinary call at $\lambda = 0.05$, the boundary oracle has complete T cap 421,318,601,825 against 435,151,104,775 for raw evaluation, a reduction of approximately 3.18 per cent. The same case improves on the informed first correction by about 76.0 per cent. This is a positive implementation result for the strike-integrated, nonnegative curvature preparation in the retained pool. Stable quadrature loses to the best second-correction implementation in every active case; its exact finite sixth moments and fixed-parameter error allowances are nevertheless supplied in full. The exact integer simplifier produces no production improvement.

The independent checks reconstruct scalar contracts, integer profiles, allocation decisions and full estimator plans, with separate small-fixture replay of dirty numerical and flag workspaces. They verify actual node rounding, retained information, clean inverse and controlled-phase conventions. Large production profiles are checked by composition, rather than exhaustive state simulation. Complete counts and setup ledgers are retained in the supporting index and result archives.

The financial certificate uses factorised finite MGFs with explicit exponential-series remain-

ders. Scalar recurrence, coefficient formation, normal quantiles, tensor-factor products and review work remain real construction costs. The previous higher-dimensional precision search and its audit took about 22.71 and 8.87 minutes in retained diagnostics. These are research setup times, not single-price execution times or conversions of T gates to CPU time.

I separately compare two discretisation certificates for the same R_2, with the same actual nodes, boundary norms, dispersion envelopes and full Gaussian tail. The first uses the positive absolute weighted fundamental-theorem-of-calculus (FTC) measure; the second uses the published centred Peano identity. This is a comparison of transport order, not of the pricing corrections R_1 and R_2. Neither opponent uses a whole-box dispersion maximum or Cartesian price enumeration. I refine only the transported coordinate into $B = 1, 8, 32, 128, 1024$ contiguous blocks, keeping other coordinates factorised. Both orders use 512-bit outward arithmetic and the same finite-moment closure.

The 0.005 allocation concerns undiscounted financial transfer, not a complete price or confidence guarantee. At every level, the second-order certificate meets that allocation in 33 of 36 cases and FTC in 32. At $B = 1024$ the second-order allowances are approximately 25.7 to 72.3 per cent smaller than FTC. Refinement from $B = 1$ tightens those second-order allowances by only 0.0702 to 0.3480 per cent and rescues none of the three two-asset failures. The previously retained sharper local certificate still handles those cases. Thus this experiment supports inexpensive aggregate residual certificates; it does not identify a new refinement principle or an additional accepted price.

Assets	p	FTC passes	Peano passes	Refinement (%)	$B = 1$ (s)	$B = 1024$ (s)
2	1	9/12	9/12	0.0702–0.0839	0.019	16.95
4	3	11/12	12/12	0.2162–0.2573	0.095	80.13
8	7	12/12	12/12	0.3250–0.3481	0.470	206.67

Table 3: Value and construction cost of transported-coordinate refinement. Pass counts concern undiscounted transfer only and are unchanged at all five levels. Refinement is the range of second-order allowance reductions from $B = 1$ to 1024, including the same tail. Each time covers geometry and both certificate orders for all twelve payoffs in that asset family, excluding common prefix setup (1.276 seconds) and serialisation. The first level forms fresh grouped coefficients; later levels reuse them. These joint incremental diagnostics are neither order-specific timings nor cold-run speed ratios. The table tests whether a tighter certificate justifies additional construction work.

Independent checks reproduce the one-block aggregate bounds, evaluate 38,313 production block transforms at 768-bit precision and check 198 positive interval quadratures. A separate 1,024-bit small-grid audit checks 1,584 hybrid polynomial moments. The retained finite-measure proof covers negative and zero transform rates, raw masses and monotone Cauchy refinement. These checks support the computed allowances; they do not establish a uniform ordering of transport orders for other residuals.

The continuous classical test now uses only the same M_2, M_3, M_4 boundary constants and exponential geometry available to transport, plus additional Gaussian moment and covariance-shifted kernel evaluations. Finite Taylor pricing through degree three has remainder radius $M_4 \mathbb{E}U^4/24$. I retain the earlier successful lower orders rather than assuming each higher order improves the bound. Table 4 shows that this bounded calculation certifies 33 complete cent prices, including all 24 higher-dimensional contracts, before preparing a grid. Those are exactly the cases which pass the aggregate R_2 transfer allocation. This is a cohort observation, not a theorem relating the 0.005 transfer allocation to the complete 0.01 price radius. Hybrid moments are not themselves continuous price coefficients; the additional Gaussian queries and kernel evaluations are charged explicitly.

The degree-three panel uses 580 shared geometry exponentials, 2,520 additional paired

Assets	Degree ≤ 1	Degree ≤ 2	Degree ≤ 3	Geometric bounds
2	4/12	8/12	9/12	0/12
4	6/12	9/12	12/12	0/12
8	8/12	10/12	12/12	0/12

Table 4: Complete classical cent certificates at unchanged model inputs. Degree columns retain the union of successful orders up to the cap; all saved outward discounted endpoints also pass the cent target. The last column combines the published geometric exercise-set lower bound with the arithmetic–geometric upper bound, and uses a derived separate-strike call-spread enclosure for caps. It is not the strongest possible capped specialisation of that method. The table identifies how much pricing work remains when the transport's model and derivative information are shared.

boundary-kernel evaluations and 4,410 density evaluations; its diagnostic is about 0.185 seconds, excluding imports and serialisation. No speed ratio is inferred against joint incremental transport timings. The independent 1,024-bit check reconstructs 180 moment enclosures, 144 coefficients, 108 remainders and all price-interval propagations. The three failures are the same two-asset $\lambda = 0.05$ cap-10, cap-25 and ordinary calls; retained validated quadrature covers them. The previous higher-degree checks remain unchanged. Strong classical methods therefore still resolve all 36 prices with zero quantum calls.

I implement the published geometric bounds of Caldana et al. (2016) from the repository-labelled accepted manuscript, including its equation (27). I write

$$\begin{aligned} \sigma_G^2 &= \operatorname{Var}(\log G), \\ a_i &= \frac{\operatorname{Cov}(\log S_i, \log G)}{\sigma_G}, \\ A_i &= w_i \mathbb{E} S_i. \end{aligned} \tag{44}$$

The lower-bound objective has derivative $\phi(z)\{k - \sum_i A_i e^{a_i z - a_i^2/2}\}$. Strict convexity handles the two mixed-sign covariance families; positive basket weights do not imply a monotone conditional mean. The complete root and endpoint proof, exact brackets and cap subtraction are retained outside the main argument. This implemented enclosure meets no cent allocation here, which does not defeat the published method's stronger variants or its use as a control. Correctness and literature priority remain separate from this numerical finding.

I test the observed admission coincidence beyond the original cohort by multiplying each outer loading by $c \in \{1, 3/2, 2, 3\}$ and recomputing its risk-neutral drift. The common λ, weights, carry, contracts and numerical widths remain fixed. This gives 144 contracts. Exact scaling gives $d_i(c) = c^2 d_i(1)$, $s_i(c) = c s_i(1)$ and $R_G(c) = c R_G(1)$; every asset forward remains $100e^{0.01}$. The supporting proof derives these identities and the unchanged pricing conditions. For $V_R = \sum_i w_i \|s_i(1)\|^2 > 0$, Jensen also gives $\mathbb{E} U_c^k \geq \{e^{c^2 V_R/2} - 1\}^k$. Thus the raw fixed-degree Taylor radius can grow without the price range becoming unbounded. This does not prove difficulty for all classical pricing methods.

The fixed panel admits 81 transfer allowances and 71 strengthened Taylor-price intervals through degree three. Ten cases separate those criteria; targeted degree five resolves six and leaves four. I intersect all available lower orders and intrinsic payoff ranges before classifying a price failure. This opponent strengthening follows the first executed run and is recorded transparently; its panel and financial targets are unchanged. Earlier executions and all adverse cases remain retained. The four survivors are capped calls with $H = 1$: (n, λ, c) equals $(2, 0.4, 3)$, $(4, 0.05, 3/2)$, $(4, 0.2, 3)$ and $(8, 0.2, 3)$. Their transfer allowances are approximately 0.00342842, 0.00270704, 0.00466658 and 0.00163955, rounded upwards, while their best strengthened degree-five price radii are approximately 0.0279505, 0.0305001, 0.0509043 and 0.0127283, also rounded upwards. This establishes useful transfer certification beyond this declared finite Taylor screen.

It establishes neither complete quantum prices for these stresses nor failure of stronger classical algorithms. The original quantum resource caps are not extrapolated to altered supports or normalisers. Figure 1 shows the distinct certificate behaviour and the effect of the stronger classical check.

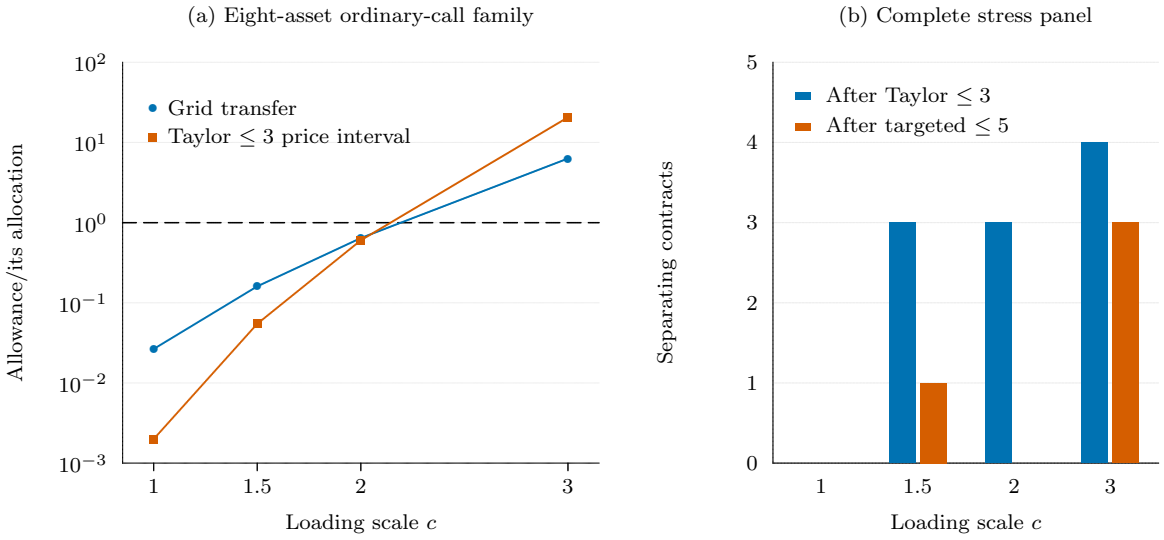


Figure 1: Financial-transfer and price certificates under a controlled dispersion stress. Panel (a) uses the eight-asset ordinary-call family at $\lambda = 0.2$, selected before outcomes. Transfer allowances are divided by 0.005 and complete Taylor-price intersection radii by 0.01; the dashed line marks admission at one. These are different error targets. Joined points guide the eye between tested scales. Panel (b) counts transfer-admitted contracts whose strengthened Taylor-price intervals fail, among all 36 contracts per scale. Degree five is evaluated only for provisional degree-three separators. Price intervals include lower-order intersections and intrinsic bounds. These are synthetic certified allowances, not observed errors, quantum prices or speed-ups.

I next price the four cap-one candidates completely. A direct conditional price certificate combines the bounded kernel with global log-basket coordinate derivatives; Appendix F derives the financial transfer and full confidence interval. I sum all 1,024 grid scores for the two-asset case. For each remaining case, a fixed 100,000-draw calculation uses the non-asymptotic inequality of Hoeffding (1963), with statistical radius 1/150 and failure allowance 1/3600. Actual 10-bit labels come from retained operating-system words; coverage explicitly assumes independent uniform words. Every evaluated score is enclosed and rounded outwards to 40-bit endpoints. The whole Gaussian tail and retained mass are included. All four complete undiscounted price radii are below 0.01.

This earlier quantum construction costs all sixteen method targets, both coherent lookup architectures and every monetary obligation. One eight-asset boundary baseline fails with deterministic allowance approximately 0.01673354; the declared scalar-address-width-14 repair passes at approximately 0.00441931. The original rejection is retained. The direct first-residual certificate is supplied to the informed R_1 opponent by taking its minimum with the older bound; no classical Monte Carlo answer is imported into that prior. The earlier first-correction caps range from approximately 25.3 to 216.1 billion logical T gates. Raw second corrections require approximately 0.301 to 2.045 trillion, and the boundary construction is not the cheapest in any of these four cases. These are conditional costs in the declared baseline and repair policy, not globally optimised resource bounds.

Independent replays verify financial enclosures, complete confidence and emitted resource charges. Full traces, rejected plans, scalar tables, source versions and reconstruction records remain in the supporting archive. The reconstruction protocol precedes fresh execution with

Assets	λ	c	Classical method	Undiscounted interval	Selected R_1 T cap
2	0.4	3	Exact sum	$[0.406517, 0.406908]$	25,306,395,925
4	0.05	1.5	Conditional MC	$[0.479091, 0.496631]$	90,433,832,750
4	0.2	3	Conditional MC	$[0.437322, 0.452372]$	97,976,923,250
8	0.2	3	Conditional MC	$[0.442736, 0.460092]$	216,055,259,050

Table 5: Complete classical prices and independently constructed quantum resource caps for the four synthetic cap-one candidates. Classical interval endpoints are rounded outwards to six decimals. The two-asset exact sum is deterministic; each Monte Carlo price has failure allowance 1/3600 under the stated source premise. Quantum columns are ideal-logical resource plans, not measured prices or running times, with the same per-price failure budget and radius 0.01. The informed positive first correction is cheapest within this earlier sixteen-target quantum family. These additive clean caps precede the cancellation-aware comparison in Table 6. Exact results, financial errors, all alternatives and source hashes remain retained.

earlier expected outcomes known; no blind first-outcome registration is claimed. All retained evidence is synthetic. The complete comparison resolves the four candidates classically and supplies valid conditional quantum plans. Figure 1 identifies failures of the declared Taylor screen, rather than prices unavailable to classical calculation.

9.1 A certified factor-basis comparison

Orthogonal Gaussian reparameterisation is established in classical derivative pricing (Imai and Tan, 2006). For an exact orthogonal matrix O, replacing R by RO preserves RR^{T}, risk-neutral drifts and the continuous asset law because OY has the law of Y. It also preserves c_0, c_1 and continuous U moments. The truncated and rounded product laws generally change. With full loading maxima $A_j(O) = \max_i |(RO)_{ij}|$ and $r_j(O) = (R_G O)_j$, the direct cap-one transfer becomes

$$B_1(O) = \sum_j \{A_j(O) + |r_j(O)|\}, \tag{45}$$

$$B_2(O) = \sum_j A_j(O)^2 + \|R_G\|_2^2, \tag{46}$$

$$\beta(O) = a^{p-1}\{b_1 M_1 B_1(O) + b_2(M_1 + M_2)B_2(O)\} + (1 - a^p). \tag{47}$$

Here a is one-coordinate retained mass, b_1 is the sum of absolute cell first moments and b_2 is half the summed second moment. The earlier global derivative proof applies in the new coordinates; cell Taylor expansion and tensor telescoping give this allowance. The full invariance and finite-law proof is retained in the supporting derivation.

I register identity and each single plane with cosine 3/5 and sine $\pm 4/5$, giving $1, 7, 7, 43$ candidates across the four cases. Earlier loading diagnostics were known. The exact minimum direct allowance selects one candidate per case; I then rebuild every finite moment, support, normaliser, arithmetic certificate and complete method cost for that basis and identity. The direct bounds improve by about 25.25, 23.83 and 10.56 per cent in the three multi-factor cases, but complete selected T caps fall by only 0.05, 0.16 and 2.07 per cent. The phase sizes remain unchanged. One rotated stable-Gauss allocation fails and a boundary cost rises substantially; both remain recorded. The two-asset identity is the control. This is a restricted paired cost comparison, not cost optimisation over all 58 bases. Every representation and certificate is available to the classical opponents; their existing continuous price certificates remain valid for the invariant target.

9.2 A shared threshold pricing preparation

I develop the existing positive kernel representation into a different coherent first correction. It trades conditional-price evaluation for a binary event; the financial law and analytic control remain unchanged.

Proposition 9.1 (A positive coupled price correction)**.** *For the finite nonnegative measure ν in (8), of mass $0 < H < \infty$, let X have law ν/H and let N be an independent standard normal variable. For $\Delta \geq 0$ and $\lambda > 0$,*

$$P_\nu(m + \Delta) - P_\nu(m) = H \Pr\{m < X - \lambda N \leq m + \Delta\}. \tag{48}$$

For a capped call, $X = \log(1 + (H/K)U_0)$ with U_0 uniform on $[0, 1]$. Cash digitals and finite positive mixtures have the same representation; the prepared threshold law must separately be certified.

Conditioning on the common threshold converts each endpoint to a Gaussian half-line probability. Subtracting nested events gives the positive correction while preserving its payoff normaliser. Appendix A.8 gives the full proof.

The same auxiliary draw is essential for both endpoints. The score has normaliser H for every draw. In contrast, $\Delta P'(m + t\Delta)$ is bounded by $\Delta_* M_1$; clipping that integrand at H can change its expectation. Its retained allowance screen does not constitute a fully costed rival.

For the cap-one implementation I use 10-bit inner normal labels, four-bit uniform strike labels and a six-bit strike-log table. Outer scalar addresses remain at twelve bits. Let $\mathcal{L}$ be this finite set of labels, with $|\mathcal{L}| = 2^{10p+14}$. The clean reversible preparation is

$$A_{\mathrm{ev}}|0\rangle = |\mathcal{L}|^{-1/2} \sum_{\ell \in \mathcal{L}} |\ell\rangle |b_\ell\rangle |0_{\mathrm{work}}\rangle, \tag{49}$$

$$b_\ell = \mathbf{1}\{\widehat{L}_\ell < \widehat{Z}_\ell \leq \widehat{R}_\ell\}. \tag{50}$$

Here $\widehat{L}_\ell, \widehat{R}_\ell, \widehat{Z}_\ell$ are the actual represented lower, upper and threshold registers with a common exact offset J. Removing J gives the financial endpoints and threshold. Two ordered comparisons and their inverses implement the interval flag; all private work is cleaned before reflection. The represented upper endpoint is the same lower register plus a nonnegative log-ratio, with range guards excluding overflow. Canonical estimation uses this actual flag and its actual reflected core.

The added approximation is explicit. With $M_S = 16$, $M_N = 1024$, $a_N = \Phi(6) - \Phi(-6)$ and normal node displacement δ_N, take checked $\nu_0 \geq \sup \phi$ and $\nu_1 \geq \sup |\phi'|$ and put

$$\beta_S/H = \frac{(H/K)^2}{12 M_S^2}(\nu_1/\lambda^2 + \nu_0/\lambda), \tag{51}$$

$$D_N = (1 - a_N)/2 + a_N/(2M_N) + \nu_0 \delta_N, \tag{52}$$

$$\eta = \min\{1, \beta_S/H + 2D_N + \nu_0(e_\ell + e_u + 2e_z)/\lambda\}. \tag{53}$$

The e terms bound actual endpoint and threshold arithmetic. A midpoint Taylor bound proves β_S; a labelled quantile coupling bounds both CDF conventions by D_N. Expanding and shrinking event endpoints against the continuous Gaussian proves the arithmetic allowance, including empty intervals and discrete ties. Finite atoms are not treated as continuous points. The supporting proof gives every derivation and circuit invariant. The outer transfer applies to the smooth conditional correction, and the uniform η then relates its nodal values to expected binary flags. No derivative of a discontinuous flag is assumed. Only the outer mass is decoded: inner normal truncation is already in D_N. If $\bar{a}_p$ bounds the outer mass and r_a is its radius, then on the certified estimation event

$$|\widehat{V} - V| \leq r_{c_0} + \beta_y + \bar{a}_p H(\eta + \varepsilon_{\mathrm{AE}}) + r_a H. \tag{54}$$

The complete plan selects $\varepsilon_{\mathrm{AE}}$ to meet radius 0.01 with failure allowance $1/3600$.

I test the encoding at four already selected bases, with no repairs. I then strengthen the numerical competitor by independently allowing $b_A, b_G \in \{6, 8, 10, 12\}$ for its two price tables, leaving other widths fixed. This registered 64-configuration pool retains twelve rejected budgets. Every admitted cost includes the same full financial and confidence obligations. Its numerical minima use $b_A = b_G = 10$. Figure 2 exposes that numerical pool.

I also cost a direct common-normal rival. With $f(x) = \min\{1, 100(e^x - 1)_+\}$ and $\Delta = \log T \geq 0$, its score is

$$h_{m,\Delta}(N) = f(m + \Delta + \lambda N) - f(m + \lambda N) \in [0, 1]. \tag{55}$$

Both payoffs use the same normal label. Exact monotone clamps restrict exponential evaluation to $[0, 1/64]$, since $\log(101/100) < 1/64$. The power-series inequality $e^x \leq (1 - x)^{-1}$ gives the rational output bound $64/63$. I retain the initial broad-headroom, rational-range and common-width studies. I then register independent geometric and arithmetic exponential addresses $b_G, b_A \in \{8, 9, 10, 11, 12\}$ with normal widths $\{8, 10, 12\}$. This gives 300 settings: 48 exact diagonal references and 252 fresh graphs, including the previously untested diagonal width nine. All pointwise and smoothing budgets are admitted under the same conservative certificate policy. Both lookup architectures and all outcomes remain retained.

Individually monotone approximations need not preserve their mutual order. I retain the final projection in the actual probability score:

$$\begin{aligned} \widehat{h}_y(n) &= \mathrm{clip}_{[0,1]}\big(\widehat{f}_A(\widehat{X}_A) - \widehat{f}_G(\widehat{X}_G)\big), \\ \mathrm{clip}_{[0,1]}(v) &= \min\{1, \max\{0, v\}\}. \end{aligned} \tag{56}$$

An independently checked scalar-domain witness gives a negative represented difference at equal arguments. The guarded positive subtraction masks this modular difference before it becomes a financial value, and its inverse cleans the workspace. Projection is 1-Lipschitz and fixes the ordered ideal payoff difference, so it preserves the charged approximation bound. The full proof and exact witness are retained in the supporting appendix; the witness does not assert occurrence in the four financial grids.

A pointwise payoff slope can overcharge shared endpoint errors. I integrate the continuous normal before differentiating. For a positive threshold measure μ of mass one, define

$$\begin{aligned} P_{\lambda,\mu}(m) &= \int \Phi((m - a)/\lambda)\, \mu(da), \\ R_{\lambda,\mu}(m, \Delta) &= P_{\lambda,\mu}(m + \Delta) - P_{\lambda,\mu}(m), \\ \kappa_0 &= \nu_0/\lambda, \qquad C_\lambda(d) = \min\{\kappa_0, \nu_1 d/\lambda^2\}, \qquad d \geq 0. \end{aligned} \tag{57}$$

Here ν_j bounds the unit Gaussian density derivative of order j. Positivity gives $0 \leq P' \leq \kappa_0$ and Gaussian smoothing gives $|P''| \leq \nu_1/\lambda^2$. For a true gap $0 \leq \Delta \leq \Delta_*$, a common fixed shift of size E_m and a gap error e_Δ therefore satisfy

$$|R_{\lambda,\mu}(\widehat{m}, \widehat{\Delta}) - R_{\lambda,\mu}(m, \Delta)| \leq C_\lambda(\Delta_*) E_m + \kappa_0 e_\Delta. \tag{58}$$

The critical step perturbs the gap at fixed $\widehat{m}$, then shifts both endpoints at the true gap. A fixed strike, positive strike mixture and the capped payoff derivative measure satisfy the same argument. For raw payoffs, $f' \in [0, 101]$ also makes the difference $f(x + \widehat{\Delta}) - f(x)$ 101-Lipschitz. With intrinsic exponential errors $e_{\exp,G}, e_{\exp,A}$, common inner scale error E_s, outer shift error E_m and inner-law CDF discrepancy D_N, let η_{point} denote the retained valid pointwise conditional bound and η_{previous} the minimum of previously valid conditional smoothing bounds for the same graph and target. The new raw rule is

$$\begin{aligned} \eta_{\mathrm{raw}} = \min\{\eta_{\mathrm{point}}, \eta_{\mathrm{previous}}, 1, 100(e_{\exp,G} + e_{\exp,A}) + 101 E_s + 2 D_N \\ + C_\lambda(\Delta_*) E_m + \kappa_0 e_\Delta\}. \end{aligned} \tag{59}$$

The scaling remainder depends on the normal label and retains a pointwise allowance. Treating it as a fixed shift under Gaussian smoothing is invalid. The finite-to-continuous transfer precedes smoothing, retains atomic multiplicities and charges normal rounding once. Projection cannot enlarge the separate scalar or shared-argument errors.

I sharpen the positive-measure component bound without supplying a price answer. Put $r_t(x) = \Phi(x + t/2) - \Phi(x - t/2)$ and define

$$\begin{aligned} k(t) &= \sup_x |\phi(x+t) - \phi(x)|, \qquad \mathfrak{a}(t) = 2\Phi(t/2) - 1, \\ \Omega(t,e) &= \sup_{x,\,|h|\le e} |r_t(x+h) - r_t(x)|, \quad t, e \ge 0. \end{aligned} \tag{60}$$

The sharp derivative coefficient is $k(d/\lambda)/\lambda$. The same two-segment path yields the finite bound

$$|R_{\lambda,\mu}(\widehat{m}, \widehat{\Delta}) - R_{\lambda,\mu}(m, \Delta)| \le \Omega(\Delta_*/\lambda, E_m/\lambda) + \mathfrak{a}(e_\Delta/\lambda). \tag{61}$$

Atomic measures attain each component extremum; their sum need not be jointly sharp. Unique scalar roots certify these extrema outward. The full proof also derives a stricter cap-specific envelope, with a uniform gap certificate rather than assumed monotonicity. Interval checks show derivative-component improvements of 7.86 and 16.47 per cent at $d/\lambda = 1$ and 1.5. A cap-specific example improves that component by at least 54.70 per cent. These are analytic certificate improvements, not T-gate percentages, and are not selected in the registered resource comparison below. Exact kernel-translation sensitivity has close published antecedents (Duchi et al., 2012); its consulted author text concerns an L^1 gradient bound, whereas this extremum uses a positive measure and an L^∞ kernel difference. Sharpness in this class establishes no literature priority.

I register the applicable refinement for all 348 raw, 64 numerical and four threshold configurations, using each graph's actual rounding terms. The numerical formula requires inactive input clamps and its exactly represented unit normaliser; both premises are checked. The threshold formula conditions on the strike label and separately sandwiches the normal-dependent scaling remainder. Full proofs and all three formulae are retained in the supporting appendix. All 416 certificates tighten, but selected widths, phase resolutions, repetitions and complete canonical caps are unchanged. The true gap obeys $\Delta \le U \le U_*$, giving the shared price-independent choice $\Delta_* = U_*$ and $C_\lambda(U_*) = \kappa_0$ in all four cases. A first-moment variation refinement supplies no additional gain.

The raw minima use $b_G = b_A = 9$; the numerical minima use width ten. All earlier configurations and outcomes remain retained as references.

Under the fixed graph, word precision, normal widths, unary emitters and canonical controller, an optimistic charge floor now excludes either payoff table at widths fifteen through twenty-two, irrespective of the other feasible width. Each side contributes four unary lookups in the complete preparation, with selector charge proportional to $2^b + b$. The supporting proof derives the exact count and excludes both zero-shot prior decisions. Lower omitted widths, unequal widths thirteen/fourteen, shared arithmetic and other estimators remain open. The common-information amendment preserves these circuit profiles and the selected canonical costs.

I next address the published optimisation missing from additive template charges. In operator order, let $A = U^\dagger P U \mathsf{H}$ be the clean preparation, where H prepares the input labels, U computes the guarded graph and P XORs its predicate into the flag. Since U does not act on that flag, the alternative $B = PU\mathsf{H} = UA$ has identical good probability. The same actual zero-reflection tape gives conjugate Grover operators. Its inverse restores zero work before every reflection, including the borrowed ladder scratch. Dirty terminal work is measured and reset on every allocated wire. The controlled tape has its own clean-work contract; it is not assumed to implement the ideal reflection on arbitrary dirty scratch. The supporting proof verifies both domains, the conditional global minus and unchanged QPE distributions.

Writing $F_U = T(U)$, the globally cancellation-optimised clean QPE network is the B network followed by one terminal $U^\dagger$. For R repetitions and phase size M,

$$T_{\text{clean,opt}} = T_B + RF_U, \qquad T_{\text{clean,add}} - T_B = R(2M-1)F_U. \tag{62}$$

Thus the large additive-template reduction mainly reflects established cancellation. Removing terminal cleanup saves only 0.097 to 0.195 per cent against the stronger clean comparator across the twelve method/case minima. Both variants are registered for every admitted configuration and both architectures. No gate is assigned a free cost, and no fresh whole-circuit emission is claimed: exact retained primitive profiles are transformed and their full compositions independently checked.

Assets	λ	c	Rival T cap	Event T cap	Saving (%)
2	0.4	3	11,630,271,225	10,467,507,700	10.00
4	0.05	1.5	44,122,469,300	41,861,923,075	5.12
4	0.2	3	46,988,300,100	44,709,494,725	4.85
8	0.2	3	104,186,984,850	101,750,688,975	2.34

Table 6: Complete canonical-QPE caps after the same cancellation-aware preparation is applied to every admitted method and architecture. Each rival is the cheaper deferred numerical or raw-payoff preparation. Raw payoff supplies the first, third and fourth comparator rows; numerical conditional pricing supplies the second. Every alternative has the same model, basis, analytic mean information, monetary radius 0.01 and failure allowance 1/3600. Savings are rounded to two decimals; exact costs and ratios are retained. These are sufficient logical implementation caps under synthetic contracts and finite policies, not global precision optima, quantum price observations or physical runtimes.

Against the strongest equally optimised implemented rivals, the event representation reduces complete caps by 2.34 to 10.00 per cent. The selected phase sizes and 25 repetitions are unchanged. Independent checks reconstruct 808 deferred profiles, 1,616 phase plans and 3,232 canonical or half-confidence variant charges, retaining all twelve rejected budgets. The financial certificates, priors and controller parameters are preserved exactly by conjugacy. Earlier additive costs remain historical. Compatible QSP evaluators, additional arithmetic optimisations and broader estimator families remain separate comparisons.

The broader identity is not an efficient-state-preparation theorem for every ν, and does not cover negative measures, infinite payoff mass or negative basket weights. A classical opponent integrates both auxiliaries to recover the smooth conditional score $F(Y)$; explicitly,

$$\operatorname{Var}(H\mathbf{1}_{\text{event}}) = \operatorname{Var}(F(Y)) + \mathbb{E}[F(Y)(H - F(Y))]. \tag{63}$$

The unconditioned classical score therefore has no variance improvement. For these cap-one targets, all three encodings have the same ideal global probability $\mathbb{E}F/H$ and Grover angle before discretisation. The encoder does not improve the generic query law. The complete classical prices remain valid, and no practical quantum advantage or priority for the established threshold identity is inferred.

9.3 Representation transfer across reflection samplers

For a coherent preparation A_i with a proved transported invariant, let a_i count its logical T gates and s_i those of its invariant input-zero reflection. The complete adjoint has the same T cost. The good reflection is Clifford, and all wire measurements and conditional Pauli resets retain their separately charged zero-T logical cost. The result concerns additive retained templates, with no cancellations across their boundaries.

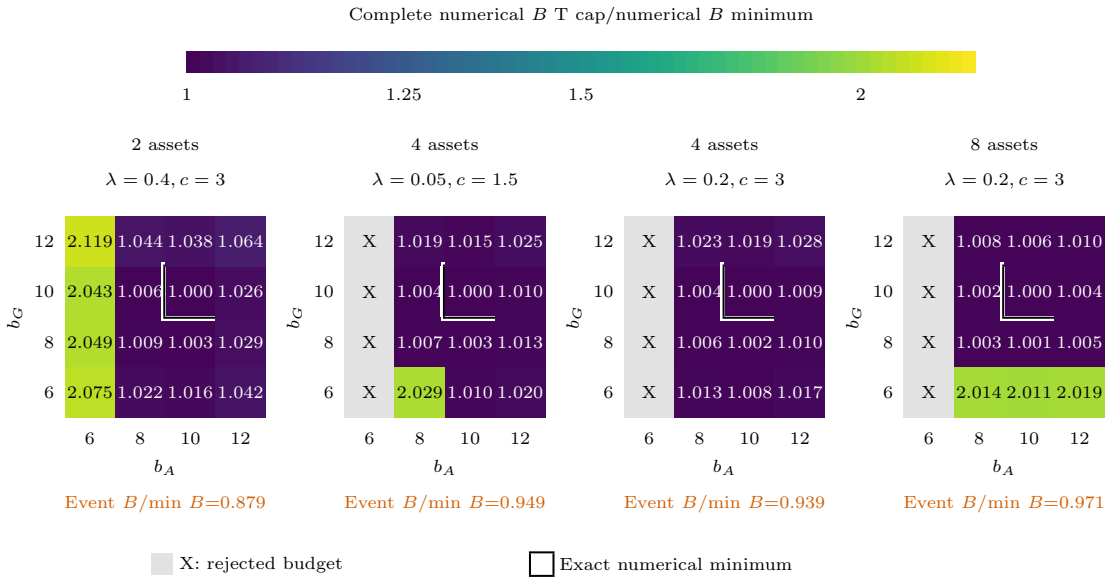


Figure 2: Complete T caps for the deferred-cleanup preparation B over the 64 registered numerical-rival configurations. Panels give asset count, common volatility λ and loading multiplier c. Columns and rows give the address widths b_A and b_G of $P(m_A)$ and $P(m_G)$; other widths and inputs are fixed. Each cell uses the cheaper complete B cap across the two retained coherent architectures. Colours and labels show its cap/the numerical B minimum in that panel, using the same shared logarithmic colour scale. Labels have three decimals; exact costs and ratios are retained. Grey crosses mark the same twelve rejected monetary budgets. Outlines identify the numerical minima, $b_A = b_G = 10$. Orange footers compare the shared-threshold B encoding with these numerical B minima; Table 6 also includes the raw-payoff pool. The same established cancellation is applied to every representation, with each configuration retaining its financial certificate, information prior and phase plan. All alternatives use monetary radius 0.01 and per-price failure allowance 1/3600. The figure displays deterministic logical resource caps for synthetic models.

Lemma 9.2 (Matched-degree preparation dominance)**.** *At every positive integer degree, the parity-specific reflection sampler has complete per-shot T cost*

$$t_i(d) = \begin{cases} da_i + (d-1)s_i/2, & d \text{ odd}, \\ da_i + (d/2-1)s_i, & d \text{ even}. \end{cases} \tag{64}$$

Candidate c has no larger cost at every degree if and only if

$$a_c \le a_b, \qquad 2a_c + s_c \le 2a_b + s_b. \tag{65}$$

For a rival with $a_b > 0$, the sharp uniform cost ratio satisfies

$$\frac{t_c(d)}{t_b(d)} \le \max\left\{\frac{a_c}{a_b}, \frac{2a_c + s_c}{2a_b + s_b}\right\}. \tag{66}$$

This is an implementation corollary of the published reflection samplers (Suzuki et al., 2020; Rall and Fuller, 2023). The structural step is that odd costs are $a_i + k(2a_i + s_i)$ and even costs are $2a_i + (k-1)(2a_i + s_i)$. Two affine inequalities therefore control every degree, including degrees outside the sampled panel. Appendix A.1 proves necessity, sufficiency, the sharp ratio and the conditional same-history transfer.

The cancellation-aware threshold preparation satisfies both inequalities strictly against each selected raw and numerical rival in all four cases. Its per-shot saving therefore extends to

the retained reflection-sampling family. This is a project-specific resource consequence of the checked pricing representation. Different rounding certificates permit different probabilities, tolerances and adaptive histories. Identical implemented probabilities and controller information are additional premises for transferring the saving to entire adaptive histories.

9.4 A matched bounded Chebyshev comparison

I give each of the three cancellation-aware selected preparations the same declared Chebyshev controller: at most twelve fresh 100-shot looks, degrees through 2,047, half the failure allowance for the pilot and half for a fresh canonical fallback. Its complete protected cap is $T_F + \lfloor T_F/10 \rfloor$. A separately registered full-pool comparison rebuilds all 808 admitted method/architecture fallbacks, preserving the twelve rejected numerical budgets. Every fallback keeps its phase size, Fourier tape and per-repetition charge, while repetitions increase from 25 to 27. Thus its protected-cap function is

$$\mathcal{C}(T_0) = \frac{27}{25}T_0 + \left\lfloor \frac{27}{250}T_0 \right\rfloor, \tag{67}$$

which is strictly increasing on the admitted canonical caps. The same precision minima therefore minimise these certified protected caps across the full pool. Their relative ordering survives this controller change, although each protected cap is approximately 1.188 times its own canonical cap. The modern branch therefore improves no worst-history guarantee against canonical QPE here.

The diagnostic uses five stipulated probabilities in the common certified ideal-mean interval and twenty paired seeds, giving 400 paired histories and 1,200 alternative histories. Shared words remove method-relative mesh and seed differences. They do not assert equality of the actual rounded financial flags. Of these alternatives, 254 pilots certify over 1,968 confidence looks. Every median exceeds its own canonical cap. The numerical competitor has the lower descriptive mean in every case, even though its matched-degree shots cost more. Its distinct financial allowances change its admissible degrees and stopping decisions. Table 7 displays this limitation beside the positive resource transfer. The supporting proof derives conditional coverage, monetary matching, pairing, discarded-pilot charges and the exact break-even criterion. Mesh means identify no expected cost at an unknown financial flag. Unchanged published ChebAE heuristics, other estimators and optimised upstream evaluators remain separate comparisons.

Assets	λ	c	Shot floor (%)	Protected saving (%)	Mesh mean ratio
2	0.4	3	9.98	10.00	1.088
4	0.05	1.5	5.11	5.12	1.025
4	0.2	3	4.84	4.85	1.088
8	0.2	3	2.33	2.34	1.054

Table 7: Cancellation-aware estimator transfer and its limit. The shot floor bounds threshold savings against both selected rivals at every positive integer degree and is rounded down. Protected savings use the full-pool minimum certified caps and the cheaper equally optimised rival. The final column gives threshold/numerical mean charge on the same stipulated mesh and paired seeds; ratios above one favour numerical pricing. These are synthetic resource comparisons, not expected-runtime or price estimates.

The transport theorem applies beyond one payoff and asset count. It certifies Gaussian integration whenever residual derivatives have finite exponential-polynomial envelopes, including conditioned classical grids and coherent financial preparation. For these baskets, moments through degree four suffice for the second-residual transfer. Their construction is polynomial in the number of exponential terms, with explicit scalar setup and tensor-factor products; the error itself is not dimension-free. The abstract finite-transform reduction shows that scalar

factorisation and Cauchy closure follow from established positive-measure tools. The residual derivative calculation is a necessary financial specialisation, with practical bounds, rather than a new general integration principle. The checked argument supplies a reusable financial certificate and reconstructed logical resource evidence. It establishes neither superiority over classical pricing nor global precision optimality. Compatible estimator comparisons beyond this bounded family and the four stressed contracts, and broader market validation, remain necessary for broader claims. The supporting index maps complete derivations, source versions, failures and byte-exact earlier archives and semantically reconstructible new evidence without expanding the main argument.

A Detailed proofs and standard derivations

A.1 Proof and scope of the reflection-sampler transfer

Proof. Put $Q_i = -A_i S_{0,i} A_i^\dagger S_g$. The unconditional global minus can be omitted because it changes a terminal shot only by a global phase. It cannot be omitted from a controlled-QPE comparison. For $d = 2k+1$, the chronological sampler prepares A_i and then applies Q_i exactly k times. It uses $k+1$ preparations, k adjoints, k zero reflections and k Clifford good reflections. For $d = 2k$ with $k \geq 1$, it prepares A_i, applies Q_i^{k-1}, applies S_g and then $A_i^\dagger$, observing the invariant input-zero event. It uses k preparations, k adjoints, $k-1$ zero reflections and k Clifford good reflections. Complete measurement and reset are appended to both constructions. Their declared T charge is zero, so these counts prove (64).

Let $q_i = 2a_i + s_i$, $\delta a = a_c - a_b$ and $\delta q = q_c - q_b$. The odd costs are $a_i + kq_i$ for $k \geq 0$ and the even costs are $2a_i + (k-1)q_i$ for $k \geq 1$. If $\delta a \leq 0$ and $\delta q \leq 0$, subtracting these formulae gives nonpositive differences in both parities. Conversely, degree one requires $\delta a \leq 0$. If $\delta q > 0$, the odd difference $\delta a + k\delta q$ is positive at a sufficiently large integer k, contradicting all-degree dominance. This proves necessity and sufficiency. At $d = 1$ strict dominance requires $\delta a < 0$; together with $\delta q \leq 0$ this is also sufficient in both parities.

For odd degrees the ratio of costs is $(a_c + kq_c)/(a_b + kq_b)$. It is a convex combination of a_c/a_b and q_c/q_b, with nonnegative weights $a_b/(a_b + kq_b)$ and $kq_b/(a_b + kq_b)$. For even degrees replace a_i by $2a_i$ and k by $k-1$; the two endpoint ratios are unchanged. A convex combination is bounded by the larger endpoint. Degrees one and two attain a_c/a_b. Along either parity the ratio tends to q_c/q_b as its degree tends to infinity, proving the final uniform sharpness statement. □

Corollary A.1 (Same-history adaptive sampler comparison)**.** *Suppose two clean representations have the same implemented good probability p, and an estimator uses the same initial prior, tolerance, confidence allocation and deterministic history-dependent degree/batch policy for both. Under ideal logical operation their complete reflection shots can be coupled to have identical observed Bernoulli outcomes, chosen degrees, batch sizes and stopping decisions. If* (65) *holds, every coupled history has no larger candidate reflection-shot T cost. If* (66) *has right side r, then the total candidate reflection-shot cost on every history is at most r times the corresponding rival reflection-shot cost. Consequently the same comparison holds for expectations whenever they exist.*

Proof. The two-dimensional reflection algebra gives the same Chebyshev Bernoulli parameter $q_d(p)$ for both clean representations at each degree. This includes $p = 0, 1$ by direct endpoint evaluation, without normalising a zero-probability basis component. At each shot, use a fresh common uniform random variable V to define both observed bits as $\mathbf{1}_{\{V \leq q_d(p)\}}$. Inductively, identical preceding histories imply the same next degree and batch size, which imply identical next observation under this coupling. The deterministic policy then makes the same next decision. Apply the pointwise per-shot inequality to each degree in the common history and sum its nonnegative costs. Taking expectation preserves the inequality. For unbounded stopping

histories, apply the same argument to finite prefixes and then use monotone convergence. Any additional fallback or non-reflection primitive requires its own cost comparison; it is not included by this corollary merely because it belongs to the estimator. □

A.2 Proof of Theorem 3.1

Proof. The identity $\phi^{(j)} = (-1)^j \operatorname{He}_j \phi$ follows by induction from $\phi' = -z\phi$ and the recurrence. Its boundedness and finite measure mass justify dominated differentiation under ν. If $D = \partial_m$, differentiating $t^{-r}Q(D)P(m + \log t)$ gives $t^{-r-1}(D - r)Q(D)P(m + \log t)$. Induction therefore yields $g_m^{(r)} = t^{-r} \sum_\ell s(r, \ell)P_\nu^{(\ell)}$ and then the integral in (10). Positivity of ν gives the bound. Since the leading term of $A_{r,\lambda}$ is $(-1)^{r-1}\lambda^{-r}z^{r-1}$, the polynomial is nonzero. Continuity and Gaussian decay give a maximiser z_*. Take $t = 1$, $m = x_* + \lambda z_*$ and $\nu = H\delta_{x_*}$. The absolute derivative is exactly HD_r, proving uniform sharpness. □

A.3 Proof of Theorem 4.1

Proof. The integral Taylor remainder is $k!^{-1} \int_1^T (T - t)^k g_m^{(k+1)}(t)dt$. Since $T \geq 1$, Theorem 3.1 bounds its absolute value by $H\overline{D}_{k+1}U^{k+1}/(k + 1)!$. Every Gaussian exponential has all finite moments; finite sums preserve this property, so the remainder is integrable. Taking expectations gives (17). The signed-Stirling derivative formula, (11) and (12) evaluate all its coefficients. The finite collection of exponential and scalar normal functions is computable at a positive scale, and outward arithmetic preserves their enclosures. □

A.4 Proof of Theorem 5.1

Proof. Linear growth gives an integrable payoff. For a single call, put $x = m - \log(K_j/S_{\text{ref}})$. Completing the Gaussian square gives $P_j = K_j\{e^{x+\lambda^2/2}\Phi((x + \lambda^2)/\lambda) - \Phi(x/\lambda)\}$. The identity $e^{x+\lambda^2/2}\phi((x + \lambda^2)/\lambda) = \phi(x/\lambda)$ gives $P_j' = K_j e^{x+\lambda^2/2}\Phi((x + \lambda^2)/\lambda)$ and $P_j'' - P_j' = K_j\phi(x/\lambda)/\lambda$. Gaussian density times an exponential and a polynomial dominates every required derivative on a compact m interval, justifying differentiation. The affine terms are killed by $D(D - 1)$; finite linearity proves (21). The chain rule gives $g_m''(t) = t^{-2}C_2(m + \log t)$ for $g_m(t) = P_F(m + \log t)$. Integrating g_m'' twice proves (20). Its measure and U^2 are nonnegative, and $t \geq 1$, proving both bounds. □

A.5 Proof of Corollary 5.2

Proof. Theorem 5.1 bounds the exact residual between zero and $M_2U^2/2$ pointwise on the full Gaussian law. All lognormal ratio moments are finite, so taking expectations proves both endpoints. Gaussian tilts evaluate c_0, c_1 and the continuous moment in finite sums. Outward coefficient errors enlarge the endpoints when they are implemented. □

A.6 Proof of Proposition 5.3

Proof. Equation (20) gives the conditional mean. The finite support makes second moments finite. Subtract the conditional mean and expand the square; the mixed term has conditional expectation zero. □

A.7 Proof of Theorem 6.1

Proof. Insert the exact coefficient, mass-weighted finite financial correction, finite auxiliary expectation and implemented expectation in succession. Their deterministic discrepancies are bounded by the stated terms. The decoder is $B(2p - 1)$ or Bp, giving $dB\eta_p$. Write the external-factor discrepancy against the implemented bounded score, giving $e_F B$. The triangle inequality

proves the bound on the estimator success event. Exact integer signed-part recombination preserves the centred observable, so its deterministic bound is the centred one. □

A.8 Proof of Proposition 9.1

Proof. Conditioning on $X = x$ gives $\Pr\{x - \lambda N \le m\} = \Phi((m - x)/\lambda)$. Integrating against ν/H and subtracting the two nested half-line events proves the identity. Gaussian convolution removes continuous endpoint atoms. The cap law follows from the threshold payoff integral and the change of variables $x = \log(1 + (H/K)u)$. □

A.9 Equivalent pricing measure and admissible trading

The strictly positive process $L_t = \exp(-\theta \cdot W_t - \|\theta\|^2 t/2)$ is a martingale of mean one: the assertion follows by conditioning on independent Gaussian increments and evaluating their exponential moment. It defines an equivalent probability Q on the finite horizon. Multiplying an increment's normal density by its exponential factor shifts its mean by $-\theta\Delta t$, preserving covariance and independence. Therefore $W_t^Q = W_t + \theta t$ is Brownian under Q, and the stock drift becomes $r_f - q_i$.

The discounted cumulative gain

$$e^{-r_f t} S_i(t) + \int_0^t e^{-r_f u} q_i S_i(u)\, du \tag{68}$$

has zero drift under Q. The explicit GBM solution has finite second moments on $[0, \tau]$, so its stochastic integral is a true martingale. For a self-financing admissible zero-cost strategy whose discounted wealth is bounded below, localisation followed by conditional Fatou's lemma makes the discounted wealth a supermartingale. Nonnegative terminal wealth which is positive with positive probability would contradict its nonpositive expectation. This establishes absence of arbitrage in the stated admissibility class. Full row rank is sufficient, rather than necessary. If Brownian risk is unspanned, I use this stipulated Q; the quantum estimator does not select a new pricing measure.

A.10 Common-factor covariance decomposition

The equal common Gaussian loading is available for every positive-definite maturity log-covariance matrix C. Let $\lambda^2 = (\mathbf{1}^\mathsf{T} C^{-1} \mathbf{1})^{-1}$. Cauchy–Schwarz applied to $C^{1/2} v$ and $C^{-1/2}\mathbf{1}$ gives

$$v^\mathsf{T} C v \ge \lambda^2 (\mathbf{1}^\mathsf{T} v)^2, \tag{69}$$

so $C - \lambda^2 \mathbf{1}\mathbf{1}^\mathsf{T}$ is positive semidefinite. It kills $C^{-1}\mathbf{1}$ and has rank $n - 1$, because a rank-one subtraction from positive-definite C loses exactly that one direction. A factor RR^T of this residual matrix and an independent common Z reproduce C exactly. The original drifts use its unchanged diagonal variances. This established Gaussian factorisation shows the analytical model coverage; extracting a common factor is not itself a new asset law or a proof of a favourable numerical cost.

A.11 Gaussian tilt and differentiated conditioning

For (12), use $e^{v\cdot y - \|y\|^2/2} = e^{\|v\|^2/2} e^{-\|y-v\|^2/2}$ and shift the integration variable. Independent Gaussian convolution changes the common scale to σ. Bounded differentiated kernels justify passing each finite derivative through the expectation. This remains valid at $R_G = 0$.

A.12 Closed-kernel differentiation and independent verification

With the variables in (19), the density identities $e^{m+\sigma^2/2}\phi(z_0) = \phi(x_0)$ and $e^{m+\sigma^2/2}\phi(z_1) = (1 + H/K)\phi(x_1)$ cancel the boundary terms in differentiation. They give $P'_{\sigma,H} = Ke^{m+\sigma^2/2}[\Phi(z_0) - \Phi(z_1)]$; higher derivatives follow by the product rule and Hermite recurrence. The retained audit compares these evaluations with differentiated series and the strike-integral representation. A formal closed formula is not used as its own verification.

A.13 Separated Fourier and Hermite comparators

For comparison with the earlier separated bounds, Gaussian Fourier inversion gives $\mathcal{B}_j = (2\pi)^{-1}\int_{\mathbb{R}} |u|^j e^{-u^2/2}du$ and $\mathcal{H}_j \le \mathcal{B}_j$. Direct integration gives $\mathcal{B}_0 = (2\pi)^{-1/2}$, $\mathcal{B}_1 = \pi^{-1}$ and integration by parts gives $\mathcal{B}_j = (j-1)\mathcal{B}_{j-2}$. Thus the individual-norm triangle and Fourier triangle are distinct competent comparators. Retaining both separates a gain from cancellation from a gain obtained by tightening each derivative separately.

A.14 Canonical phase-grid error derivation

Equation (43) follows from the two Grover eigenphases and the nearest phase-grid outcomes: writing $p = \sin^2\theta$, the relevant phase error is at most π/M, and Taylor expansion of $\sin^2$ bounds the first-order term by $2\sqrt{p(1-p)}\pi/M$ and the second-order term by π^2/M^2. The probability of the nearest-grid event is bounded below by the retained canonical phase-estimation theorem. The exact finite confidence calculation and phase-tape verification remain in the supporting derivations.

A.15 Fourier synthesis and retained tape diagnostics

The inverse Fourier transform has explicit Clifford and T sequences, based on certified single-qubit rotation approximations (Ross and Selinger, 2016). Scalar phases are retained until they are physically irrelevant to the final measurement. For phase widths through 64, distant unsupported controlled phases are replaced by identity with a charged operator-norm error; retained approximations and omitted phases are summed by a telescoping product bound. The complete largest-width bound is about 0.0022013, below the stipulated 1/64 implementation limit. The 64-bit tape has 167,223 T gates; the earlier 32-bit profile is preserved at 68,823 T gates. The extension charges omitted distant phases rather than silently assigning them exact rotations. This is an implementation-error statement, not an execution on a quantum device.

B Complete root isolation for the joint norm

This appendix gives the finite certificate behind Theorem 3.1. Write $A = A_{r,\lambda}$ and $Q = A' - zA$. Since $(A\phi)' = Q\phi$, every nonzero absolute maximum is at a real root of Q. Zeros of A and limits at infinity contribute zero. For rational λ, all coefficients are rational and $\deg Q = r$. Replace Q by its monic square-free part $q = Q/\gcd(Q, Q')$; this keeps every distinct root.

Start a remainder sequence with $q_0 = q$, $q_1 = q'$ and define q_{j+1} as minus the polynomial remainder of q_{j-1} by q_j. Positive rescalings do not change signs. The sequence ends in a nonzero constant. Consecutive terms cannot vanish together, since the recurrence would propagate that zero to the last constant. At an internal zero the neighbouring terms have opposite signs, so their adjacent comparisons contribute one sign variation on both sides. At a root of q_0, its nonzero derivative makes the variation count decrease by one. Thus, omitting zero values, $V(a) - V(b)$ counts roots in (a, b) when a, b are nonroots.

Every root lies within $1 + \max_{j<r} |a_j/a_r|$ for the coefficients of q. Outside that radius the absolute sum of its lower terms is strictly less than the leading term, so cancellation is impossible.

Choose a larger integer radius and subdivide rationally, preserving the exact root-count sum. If a proposed midpoint is a root, choose a nonroot among the $r+1$ interior fractions $j/(r+2)$ of the interval; a degree-r polynomial cannot vanish at all of them. Every such split contracts each child by at most $(r+1)/(r+2)$. Distinct roots have positive separation, so repeated contraction first isolates each and then attains the required interval width. This proves finite termination, including rational roots.

Evaluate $|A\phi|$ with an outward Arb ball covering both exact endpoints of each isolating interval. The largest upper endpoint is a global upper certificate, because all maximising roots are covered. Evaluation at midpoints gives lower bounds on the global supremum even when those points are not roots. The independent implementation reconstructs the rational polynomials, verifies total root counts and interval endpoint signs, and uses higher-precision Gaussian evaluations to check the retained envelopes. No claim that interval midpoint evaluation is the exact maximum is needed.

C Signed interpolation and arithmetic contracts

Let $f \in C^2([a, a+s])$ and grid spacing $h = s/2^b$, with a power-of-two span s and $b \le q + \log_2 s$. At every endpoint x_j, retain an unsigned dyadic value $n_j 2^{-q}$ and an outward error e_j. On a cell $[u, v]$, the exact linear interpolation error equals $f''(\zeta)(x-u)(x-v)/2$ for some interior ζ. Subtract the linear interpolant and a multiple of $(t-u)(t-v)$ chosen to give a third zero at x, then apply Rolle's theorem twice to prove this identity. Since $(x-u)(v-x) \le h^2/4$, endpoint rounding and one integer truncation give

$$e_f \le \max_j e_j + \|f''\|_\infty h^2/8 + 2^{-q}. \tag{70}$$

An argument error η_x adds $\|f'\|_\infty \eta_x$ if both arguments remain in the declared interval. Projection onto that interval does not enlarge the distance from an admissible exact argument.

The actual table stores the base integer, the absolute difference to the next endpoint, and its decreasing bit. It multiplies the unsigned magnitude by the within-cell remainder and chooses unsigned addition or subtraction. The adjustment cannot underflow a decreasing base: it is at most the endpoint difference, and both endpoints are nonnegative. The final endpoint has its own address, so saturation at the right endpoint does not wrap to the left. The packed magnitudes, intermediate additions and input slices are checked against their actual widths. Outward endpoints are coarsened downwards and upwards to $2^{-384}\mathbb{Z}$ before serialization; the enlarged intervals determine all nodal errors. Extreme Gaussian tails are not treated as exact zeros.

For $x \ge a > 0$, usable first and second derivative bounds are a^{-1}, a^{-2} for $\log x$, and $2a^{-3}, 6a^{-4}$ for x^{-2}. For e^x both are e^{a+s}. For the boundary scalar, $C_2' = C_3 + 2C_2$ and $C_2'' = C_4 + 5C_3 + 4C_2$ give valid slope and curvature bounds, divided by $2M_2$ for the capped h kernel. I intersect them with independent two-density derivative bounds when smaller. Capped price tables receive $|P'| \le M_1$, $|P''| \le M_1 + M_2$ and $|P'''| \le M_1 + 3M_2 + M_3$, intersected with $H\lambda^{-j}\|\phi^{(j-1)}\|_\infty$. Vanilla raw-price tables use the finite-domain forward bound for P'; its boundary-density scalar remains globally bounded. These inequalities are shared across methods and never change a probability's decoder.

For products, either identity $\widehat{x}\widehat{y} - xy = x(\widehat{y}-y) + \widehat{y}(\widehat{x}-x) = y(\widehat{x}-x) + \widehat{x}(\widehat{y}-y)$ yields a valid absolute error. Taking the smaller range-based upper bound and adding one truncation unit preserves small-factor suppression. Addition and subtraction add operand errors. Scaling charges its rounded coefficient and any dropped bits. Each node independently propagates implemented ranges; intersecting them with an ideal range expanded by its certified error retains correlations without removing rounding. Empty intersections or domain violations reject the graph. Signed final differences must fit strictly within the unambiguous two's-complement range be-

fore projection. Positive-subtraction temporaries are predicate-masked before any downstream multiplication. These checks are essential hypotheses of the monetary certificate.

D Auxiliary midpoint and stable quadrature proofs

For a Beta cell centred at z_j with width h, write $u = z - z_j$. The density is $2(1 - z_j) - 2u$; direct integration gives first centred moment $-h^3/6$ and second centred moment $2(1-z_j)h^3/12$. Taylor's formula, summed over cells with their exact positive masses, therefore gives

$$|\mathbb{E}f(Z_B) - \mathbb{E}f(Z_{B,N})| \le h^2\{\|f'\|_\infty/6 + \|f''\|_\infty/24\}. \tag{71}$$

For $f_y(z) = g''_{m(y)}(1 + zU)$, the derivatives are bounded by M_3U and M_4U^2. Multiplication by $U^2/2$ and finite financial expectation prove (31).

For the stable rule the Beta moments are $2/((j+1)(j+2))$. The monic polynomial $\pi_2(z) = z^2 - 4z/5 + 1/10$ is orthogonal to $1, z$, by substitution of these moments. Its roots and positive weights are those in (26); the first four moments show exactness through degree three. Direct integration gives $\int \pi_2^2\, 2(1-z)dz = 1/300$. For $f \in C^4[0,1]$, its cubic Hermite interpolant H_f agrees with f, f' at both nodes. At any non-node z, subtract a multiple of π_2^2 chosen to create an additional zero there. Four applications of Rolle's theorem, counting the two double zeros, give $f(z) - H_f(z) = f^{(4)}(\xi_z)\pi_2(z)^2/4!$. At a node the error is zero. Positivity and exactness of the quadrature then imply error at most $\|f^{(4)}\|_\infty/7200$. Taking $f = f_y$ bounds its fourth derivative by M_6U^4. The factor $U^2/2$ gives the first term of (32). Replace nodes first under the true positive weights, using M_3U; then replace weights at the dyadic nodes, using M_2. This gives the two remaining terms after finite expectation. All nodes lie in $[0,1]$. The graph's dyadic constants are exact rational operations; their discrepancy from the ideal quadrature is charged here once.

E Gaussian transport and factorised financial error

I complete the financial transfer for the four- and eight-asset models by integrating derivative envelopes against a positive transport measure. This retains dispersion moments instead of replacing U by its largest value on $[-L, L]^p$. Let ν be the unnormalised truncated Gaussian measure, with mass a, and let $\tau = (a/N)\sum_c \delta_{c_c}$ use the actual rounded nodes in the same CDF cells. For their centred cell moments $\mu_{1,c}, \mu_{2,c}$, the weighted identity of Cerone (2002) (Theorem 3, $n = 1$, $\alpha = a_c$, $\beta = b_c$, $x = c_c$, $w = \phi$) gives

$$\int f\, d(\nu - \tau) = \sum_c \mu_{1,c} f'(c_c) + \int f''\, d\rho_2, \tag{72}$$

where ρ_2 has nonnegative densities $\int_{a_c}^{z}(z-y)\phi(y)dy$ to the left of c_c and $\int_z^{b_c}(y-z)\phi(y)dy$ to its right. Its mass is $\sum_c \mu_{2,c}/2$. For $\rho_1 = \sum_c \widehat{\mu}_{1,c}\delta_{c_c}$ with certified $\widehat{\mu}_{1,c} \ge |\mu_{1,c}|$, the absolute transfer is bounded by $\int |f'|d\rho_1 + \int |f''|d\rho_2$.

For the two-asset local variant, I keep the signed linear sum and bound the curvature on retained cell segments:

$$\beta_Y \le B_{\text{tail}} + \left|\sum_c \mu_{1,c} R'(c_c)\right| + \sum_{c,I} D_{c,I}\rho_2(I), \tag{73}$$

where $D_{c,I}$ encloses $\sup_I |R''|$. Nonnegative transport weights justify outward endpoint extensions. The full local proof retains the closed Gaussian residual derivatives, primitive weight integrals and independent automatic-differentiation checks.

The exact moment-generating function of the second transport measure is

$$\mathcal{M}_{\rho_2}(v) = \frac{e^{v^2/2}\{\Phi(L-v) - \Phi(-L-v)\} - (a/N)\sum_c e^{vc_c} - v\sum_c \mu_{1,c} e^{vc_c}}{v^2}, \qquad v \neq 0. \tag{74}$$

This follows by substituting $f(y) = e^{vy}$ into the exact identity, with continuous minus rounded measures in that order. At zero I use its mass directly. Near zero, the enclosure $e^{-|v|L}\rho_2(\mathbb{R}) \leq \mathcal{M}_{\rho_2}(v) \leq e^{|v|L}\rho_2(\mathbb{R})$ prevents cancellation in the quotient.

Define $\Lambda_{j,k} = \nu^{\otimes(j-1)} \otimes \rho_k \otimes \tau^{\otimes(p-j)}$ and $\langle h \rangle_{j,k} = \int h \, d\Lambda_{j,k}$. The product-measure telescoping identity then proves

$$\left| \int R_2 d\nu^{\otimes p} - \int R_2 d\tau^{\otimes p} \right| \leq \sum_j \{\langle |\partial_j R_2| \rangle_{j,1} + \langle |\partial_{jj} R_2| \rangle_{j,2}\}. \tag{75}$$

All raw masses remain inside the hybrid measures.

For $u = T_j$, $v = T_{jj} \geq 0$ and $r = r_j$, let $D_1 \geq \|C_2'\|_\infty$, $D_2 \geq \|C_2''\|_\infty$ and $D_{31} \geq \|C_3'\|_\infty$, in addition to the shared M_k bounds. The integral representation of R_2 implies

$$|\partial_j R_2| \leq M_2 U|u| + \tfrac{1}{2}|r| D_1 U^2 + \tfrac{1}{6} M_3 U^2 |u|, \tag{76}$$

$$|\partial_{jj} R_2| \leq M_2(u^2 + Uv) + 2|r| D_1 U|u| + \tfrac{2}{3} M_3 U u^2 + \tfrac{1}{2} r^2 D_2 U^2 \tag{77}$$

$$+ \tfrac{1}{3}|r| D_{31} U^2 |u| + \tfrac{1}{6} M_3 U^2 v + \tfrac{1}{12} M_4 U^2 u^2. \tag{78}$$

Let $E_{j,1}$ and $E_{j,2}$ denote these two nonnegative derivative envelopes. Under the stated positive-weight exponential geometry and bounded curvature derivatives, they give the following full certificate.

Theorem E.1 (Factorised Gaussian financial transfer)**.** *For actual nodes certified inside their Gaussian CDF cells,*

$$|\mathbb{E} R_2 - \alpha_1^p \mathbb{E}_N R_2| \leq \frac{M_2}{2} \mathbb{E}[U^2 \mathbf{1}_{Y \notin [-L,L]^p}] + \sum_j \{\langle E_{j,1} \rangle_{j,1} + \langle E_{j,2} \rangle_{j,2}\}. \tag{79}$$

All envelope terms are bounded by finite positive-measure moments of exponential polynomials of degree at most four, using the Cauchy bounds below. Their factorisation preserves the raw Gaussian masses.

Proof. Split the full Gaussian integral into the box and its complement. The global curvature bound controls the latter by the displayed outside moment. The positive one-coordinate representation, applied to each term of the tensor telescope, controls the former by the two derivative envelopes. Their derivation and finite-moment evaluation follow directly. To verify the coefficients, differentiate $G = t^{-2} C_2(m + \log t)$ with $t = 1 + zU$:

$$G_j = r t^{-2} C_2' + z u t^{-3} C_3, \quad G_{jj} = r^2 t^{-2} C_2'' + 2 r z u t^{-3} C_3' + z v t^{-3} C_3 + z^2 u^2 t^{-4} C_4. \tag{80}$$

Apply the product rule to $U^2 G$, use $t \geq 1$, and integrate $1 - z$, $z(1-z)$ and $z^2(1-z)$, whose integrals are $1/2$, $1/6$ and $1/12$. Cauchy's inequality gives $\langle U|u| \rangle \leq \sqrt{\langle U^2 \rangle \langle u^2 \rangle}$ and

$$\langle U^2 |u| \rangle \leq \min\{\sqrt{\langle U^2 \rangle \langle U^2 u^2 \rangle}, \sqrt{\langle U^4 \rangle \langle u^2 \rangle}\}. \tag{81}$$

These inequalities hold for finite positive measures without a mass divisor. Every remaining moment is a polynomial of degree at most four in the asset exponentials. Its signed multinomial sum factorises into scalar MGFs, using ρ_k in coordinate j, ν before it and τ after it. Thus no N^p residual values enter the certificate. The outside-box U^2 moment follows by Gaussian tilting and contributes $M_2 \mathbb{E}[U^2 \mathbf{1}_{Y \notin [-L,L]^p}]/2$.

□

For the matched FTC comparator, compact-cell Fubini applied to $f(y)-f(c)$ gives a signed kernel $\Phi(a_c)-\Phi(z)$ on the left and $\Phi(b_c)-\Phi(z)$ on the right. Its absolute density defines ρ_0. Writing $J_v(a,b)=e^{v^2/2}\{\Phi(b-v)-\Phi(a-v)\}$, integration by parts on both segments gives, for $v\neq 0$,

$$\mathcal{M}_{\rho_{0,c}}(v)=\frac{e^{vc}\{2\Phi(c)-\Phi(a_c)-\Phi(b_c)\}+J_v(c,b_c)-J_v(a_c,c)}{v}. \tag{82}$$

At zero I use $\int_{a_c}^{b_c}|y-c|\phi(y)dy$ directly. Replacing ρ_1,ρ_2 by ρ_0 and integrating $E_{j,1}$ gives its bound for the same R_2. This independently derived common-grid adaptation does not assert equivalence to the uniform-grid theorem of Herman et al. (2026).

For any transported-coordinate block partition and nonnegative moment pairs (A_D,B_D), discrete Cauchy gives $\sum_D\sqrt{A_DB_D}\le\sqrt{(\sum_D A_D)(\sum_D B_D)}$. The minimum of the two alternative Cauchy closures also cannot increase under exact refinement, since $\sum_D\min(a_D,b_D)\le\min(\sum_D a_D,\sum_D b_D)$. All envelope coefficients are nonnegative and direct moments add. Independently rounded numerical bounds are made monotone by taking the minimum of parent and summed-child upper bounds; both are valid. For M monomials, the implemented shared grouping uses $O(p^2M)$ other-coordinate products and $O(BpM)$ subsequent block dot products, with scalar transforms, rate formation and prefix preparation charged separately. Full cell transforms, interval-series remainders, refinement proofs and independent audits are retained in the supporting comparison record.

For a fair first-correction comparison, I also transfer $R_1=R_2+UP'(m)$. The extra derivatives are $uP'+rUP''$ and $vP'+2ruP''+r^2UP'''$. Capped kernels use the shared global Greek bounds; vanilla kernels use $P'\le Ke^{m+\lambda^2/2}$, $|P''|\le Ke^{m+\lambda^2/2}+M_2$ and $|P'''|\le Ke^{m+\lambda^2/2}+3M_2+M_3$. The corresponding tilted positive-measure moments also factorise. Its tail is bounded by the tilted outside moment of $Ke^{m+\lambda^2/2}U$, with the additional H tail bound for a cap. I replace the second-residual tail by this first-residual tail.

At unchanged $L=6$, $b_y=10$, $q=28$, all 24 four- and eight-asset second-correction transfer bounds pass 0.005; their largest allowance is 0.002284 rounded upwards. All 24 first-correction competitors also pass; their largest allowance is 0.004914 rounded upwards. Each global-envelope family retains three two-asset allocation failures, for which the earlier sharper local certificate remains available. The transport argument applies to arbitrary finite positive basket weights and bounded boundary derivatives, beyond these frozen models. Its numerical scope and polynomial moment-construction cost do not establish a quantum advantage or a literature-priority claim.

F Direct capped-price transfer and full confidence

For the same positive-weight model let $f(y)=P(m_A(y))$ with $m_A(y)=\log\sum_i w_ie^{\mu_i+R_i\cdot y}$ and $0\le f\le H$. Define the positive weights $\pi_i=w_ie^{\mu_i+R_i\cdot y}/\sum_k w_ke^{\mu_k+R_k\cdot y}$ and $A_j^*=\max_i|R_{ij}|$. Differentiation gives $a_j=\partial_j m_A=\sum_i\pi_iR_{ij}$ and $b_j=\partial_{jj}m_A=\sum_i\pi_iR_{ij}^2-a_j^2\ge 0$. Combining terms with $C_2=P''-P'$ yields

$$f_{jj}=P'(m_A)(a_j^2+b_j)+C_2(m_A)a_j^2,\qquad a_j^2+b_j=\sum_i\pi_iR_{ij}^2\le(A_j^*)^2. \tag{83}$$

The shared global capped-kernel norms consequently imply $|f_j|\le M_1A_j^*$ and $|f_{jj}|\le(M_1+M_2)(A_j^*)^2$ on all real coordinates. These use full asset loadings, not centred residual loadings.

For retained measure mass a, positive cell-error masses $b_k=\rho_k(\mathbb{R})$ and $F=a^p$, tensor telescoping proves

$$\beta_{\text{cell}}=a^{p-1}\sum_j\{M_1A_j^*b_1+(M_1+M_2)(A_j^*)^2b_2\},\quad V=F\mu_N+e_{\text{cell}}+e_{\text{tail}}, \tag{84}$$

where $|e_{\text{cell}}|\le\beta_{\text{cell}}$, $0\le e_{\text{tail}}\le H(1-F)$, and μ_N is the direct uniform-label mean at the actual rounded nodes. The global bounds grow quadratically with exposure scale and do not require

residual ratio moments. They are a financial specialisation of established cell and derivative tools.

For independent bounded scores, Hoeffding (1963), Theorem 2, equation (2.6), and the two-sided union in equation (1.4), give $\Pr\{|\overline{f} - \mu_N| > \epsilon\} \le 2e^{-2n\epsilon^2/H^2}$. The full supporting proof derives this from the tilted variance for both signs. I fix $n = 100000$, $H = 1$, $\epsilon = 1/150$ before new draws. Outward evaluation verifies that the failure bound is at most $1/3600$. Masked 16-bit independent uniform words give exact 10-bit labels, each with 64 preimages. Retained words verify this implementation; they do not prove physical source independence.

Validated 128-bit scores are stored as outward 40-bit integer endpoints. Exact summation gives $\overline{\ell} \le \overline{f} \le \overline{u}$. On the confidence event, put $L_N = \max(0, \overline{\ell} - \epsilon)$ and $U_N = \min(H, \overline{u} + \epsilon)$. With outward mass endpoints F_-, F_+,

$$V \in [\max(0, F_- L_N - \widehat{\beta}_{\text{cell}}),\ \min(H, F_+ U_N + \widehat{\beta}_{\text{cell}} + H(1 - F_-))]. \tag{85}$$

All financial, numerical, statistical and mass errors enter this interval; discounting is applied once afterwards. The exact finite sum replaces the sample interval and removes the statistical term for the two-asset case.

The same bounds strengthen the first-correction opponent. For the normalised weights, define $m_G(y) = \sum_i w_i(\mu_i + R_i \cdot y) = m(y)$ and $R_{Gj} = \sum_i w_i R_{ij}$. Jensen and monotonicity give $h = f - P(m_G) \in [0, H]$. Derivative subtraction bounds $|h_j|$ by $M_1(A_j^* + |R_{Gj}|)$ and $|h_{jj}|$ by $(M_1 + M_2)((A_j^*)^2 + R_{Gj}^2)$. The same cell formula and bounded tail therefore certify R_1. Taking the minimum with its earlier valid transfer is deterministic. The oracle and scalar words remain unchanged; monetary and confidence plans use this stronger shared information. Full proofs, all observed enclosures and independent replays are indexed separately.

G Global payoff-specific norm certificates

For the cap put $a = \ell/\lambda$ and $q(z) = \phi(z) - c\phi(z-a)$. On $[0, a]$, $q' = -z\phi(z) + c(z-a)\phi(z-a) < 0$, including the endpoints. Outside that interval, its stationary equation is equivalent to

$$f(z) = \log(z/(z-a)) - az - \ell + a^2/2 = 0. \tag{86}$$

The ratio is positive on each component and no sign is lost in taking logarithms. Its derivative is $-a/[z(z-a)] - a < -a$. The limits on $(-\infty, 0)$ are $+\infty, -\infty$, and on (a, ∞) they are again $+\infty, -\infty$. There is exactly one root on each component. At a root, $q(z) = -ca\phi(z-a)/z$, positive on the negative component and negative on the positive one. Gaussian tails vanish. Thus

$$\|C_2\|_\infty = \frac{K}{\lambda} \max\{q(z_-), -q(z_+)\}. \tag{87}$$

Outward rational brackets with strict endpoint signs and the monotonicity margin enclose both roots and their full stationary-value intervals. An enclosed stationary midpoint gives a lower norm bound; the larger stationary upper enclosure gives the required upper bound. Uncertainty in ℓ and a is retained throughout, rather than rounding them first.

For the capped first derivative, $P'(m) = (K/\lambda) \int_0^\ell e^x \phi((m-x)/\lambda) dx > 0$. Normalising this integrand defines a truncated normal X of mean parameter $m + \lambda^2$ and variance parameter λ^2. Differentiation gives

$$(\log P')' = (\mathbb{E}_m X - m)/\lambda^2, \qquad (\log P')'' = -1/\lambda^2 + \operatorname{Var}_m(X)/\lambda^4. \tag{88}$$

For its density p_m and mean $\bar{x}$, integration by parts yields

$$-\operatorname{Var}_m(X)/\lambda^2 = (\ell - \bar{x}) p_m(\ell) + \bar{x} p_m(0) - 1. \tag{89}$$

Both boundary terms are positive, since $0 < \bar{x} < \ell$. Therefore $\mathrm{Var}_m(X) < \lambda^2$ and P' is strictly log-concave. Its logarithmic derivative is positive at zero and negative at ℓ, so it has exactly one global maximum. Strict outward signs enclose that zero of P''; evaluation on the whole bracket certifies M_1. The supporting proof retains an explicit positive monotonicity margin.

For $r \geq 2$, define $\mathcal{P}_2 = 1$ and $\mathcal{P}_{r+1} = \lambda^{-1}(\mathcal{P}_r' - z\mathcal{P}_r) - r\mathcal{P}_r$. The chain rule proves $C_r^{\mathrm{vanilla}}(m) = K\mathcal{P}_r(m/\lambda)\phi(m/\lambda)/\lambda$; the cap subtracts its c-weighted translate by a. Exact stationary-root isolation for $\mathcal{P}_r' - z\mathcal{P}_r$ gives $N_r = \|\mathcal{P}_r\phi\|_\infty$ outwards. Valid higher cap bounds are the smaller of $H\overline{D}_r$ and $(2K + H)\overline{N}_r/\lambda$. For $r = 2$ the two-root result replaces this triangle bound. All competitors receive the same constants, and the globally unbounded vanilla P' is bounded only on its certified finite argument domain.

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